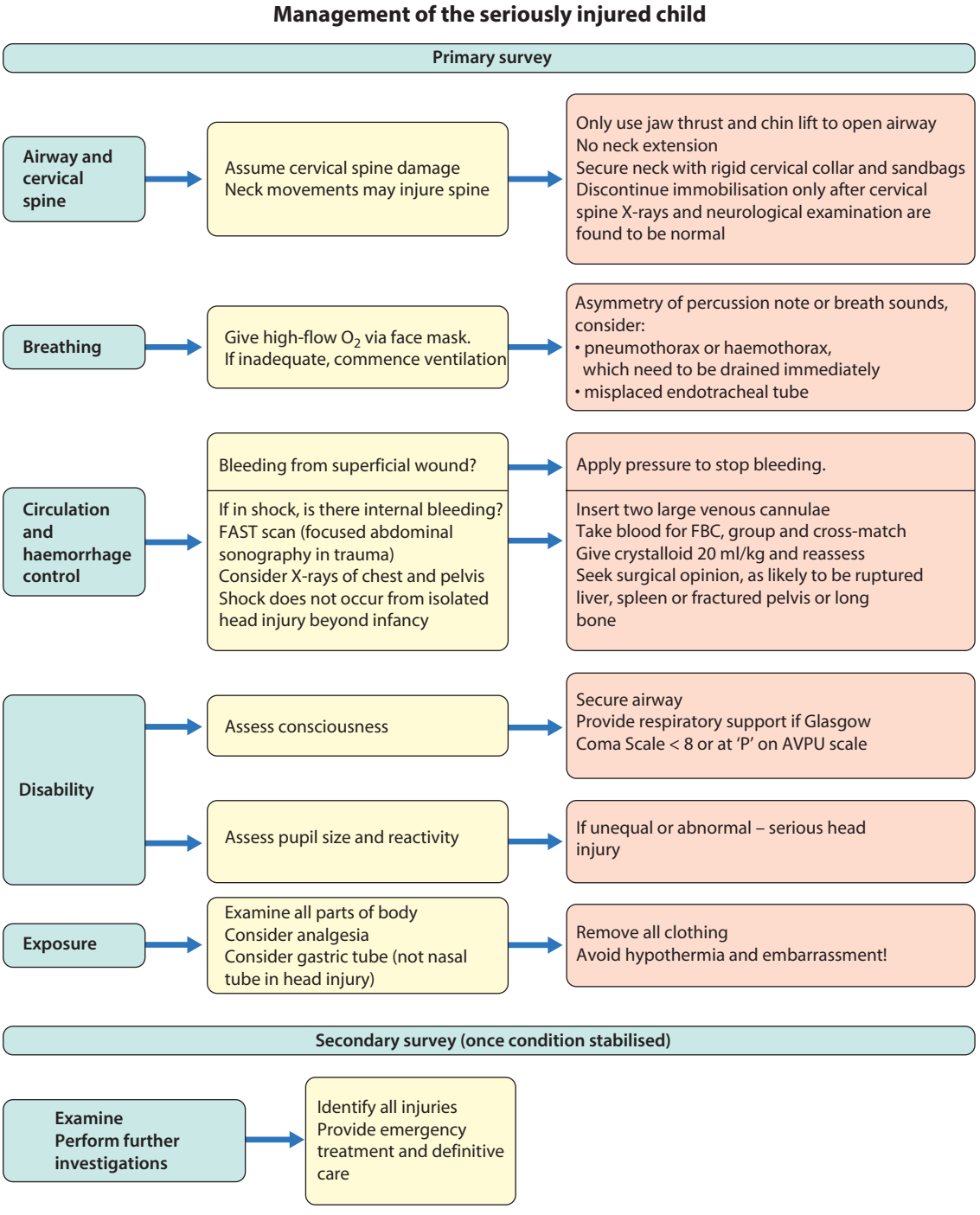


Figure 6.7 Management of the seriously injured child.

Shock

Shock is present when the circulation is inadequate to meet the demands of the tissues. Critically ill children are often in shock, usually because of hypovolaemia due to fluid loss or maldistribution of fluid, as occurs in sepsis or intestinal obstruction.

Why are children so susceptible to fluid loss?

Children normally require a much higher fluid intake per kilogram of body weight than adults (Table 6.1). This is because they have a higher surface area to volume ratio and a higher basal

Table 6.1 Fluid intake at different ages

Body weight	Fluid requirement/24 h	Volume/kg per hour (approximate)
First 10 kg	100 ml/kg	4 ml/kg
Second 10 kg	50 ml/kg	2 ml/kg
Subsequent kg	20 ml/kg	1 ml/kg
Examples of calculations		
Infant (7 kg)	700 ml	28 ml/h
Child (18 kg)	1000 + 400 = 1400 ml	40 + 16 = 56 ml/h
Adolescent (42 kg)	1000 + 500 + 440 = 1940 ml	40 + 20 + 22 = 82 ml/h

Box 6.2 Clinical signs of shock

Early (compensated)	Late (decompensated)
Tachypnoea	Acidotic (Kussmaul) breathing
Tachycardia	Bradycardia
Decreased skin turgor	Confusion/depressed cerebral state
Sunken eyes and fontanelle	Blue peripheries
Delayed capillary refill (>2 s)	Absent urine output
Mottled, pale, cold skin	Hypotension
Core–peripheral temperature gap (>4°C)	
Decreased urinary output	

metabolic rate. Children may therefore become dehydrated if:

- they are unable to take oral fluids
- there are additional fluid losses due to fever, diarrhoea or increased insensible losses (e.g. due to increased sweating or tachypnoea)
- there is loss of the normal fluid-retaining mechanisms, e.g. burns, the permeable skin of premature infants, increased urinary losses or capillary leak.

Clinical features

The clinical features of shock are manifestations of compensatory physiological mechanisms to maintain the circulation and the direct effects of poor perfusion of tissues and organs (Box 6.2).

In early, compensated shock, the blood pressure is maintained by increased heart and respiratory rate, redistribution of blood from venous reserve volume and diversion of blood flow from non-essential tissues such as the skin in the peripheries, which become cold, to the vital organs like brain and heart. In shock due to dehydration, there is usually >10% loss of body weight (see Ch. 13) and a profound metabolic acidosis which is compounded by failure to feed and drink while severely ill. After acute blood loss or redistribution of blood volume because of infection, low blood pressure is a late feature. It signifies that compensatory responses are failing.

In late or uncompensated shock, compensatory mechanisms fail, blood pressure falls and lactic acidosis increases. It is important to recognise early compensated shock, as this is reversible, in contrast to uncompensated shock, which may be irreversible.

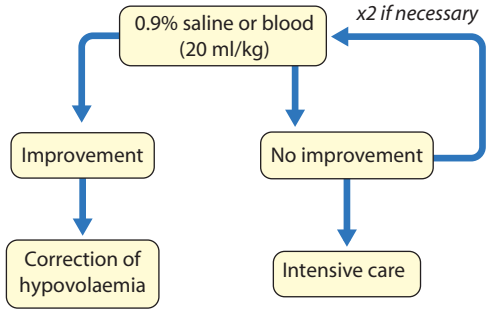


Figure 6.8 Initial fluid resuscitation in shock.

Management priorities

Fluid resuscitation

Rapid restoration of the intravascular circulating volume is the priority (Fig. 6.8). This will usually be with 0.9% saline, or blood if following trauma.

Subsequent management

If there is no improvement following fluid resuscitation or there is progression of shock and respiratory failure, a paediatric intensive care unit should be involved and transfer arranged as the child may need:

- tracheal intubation and mechanical ventilation
- invasive monitoring of blood pressure
- inotropic support
- correction of haematological, biochemical and metabolic derangements
- support for renal or liver failure.

Septicaemia

Bacteria may cause a focal infection or proliferate in the bloodstream, leading to septicaemia. In septicaemia, the host response includes the release of inflammatory cytokines and activation of endothelial cells, which may lead to septic shock. The commonest cause of septic shock in childhood is meningococcal infection, which may or may not be accompanied by meningitis. Fortunately, its incidence in the UK has fallen markedly since immunisation was introduced against meningococcal C, but other strains are still prevalent. *Pneumococcus* is the commonest organism causing bacteraemia, but it is unusual for it to cause septic shock. In neonates, the commonest causes of septicaemia are group B streptococcus or Gram-negative organisms acquired from the birth canal.

Clinical features

See Box 6.3

Management priorities

Children with septic shock need to be rapidly stabilised and may require transfer to a paediatric intensive care unit.

Antibiotics

Choice depends on the child's age and any predisposition to infection.

Fluids

Significant hypovolaemia is often present, owing to fluid maldistribution, which occurs due to the release of vasoactive mediators by host inflammatory and endothelial cells. There is loss of intravascular proteins and fluid which may occur due to the development of 'capillary leak' caused by endothelial cell dysfunction. Circulating plasma volume is lost into the interstitial fluid. Central venous pressure monitoring and urinary catheterisation may be required to guide the assessment of fluid balance. Capillary leak into the lungs causes pulmonary oedema, which may lead to respiratory failure, necessitating mechanical ventilation.

Circulatory support

Myocardial dysfunction occurs as inflammatory cytokines and circulating toxins depress myocardial contractility. Inotropic support may be required.

Disseminated intravascular coagulation (DIC)

Abnormal blood clotting causes widespread microvascular thrombosis and consumption of clotting factors. If bleeding occurs, clotting derangement should be corrected with fresh frozen plasma and platelet transfusions.

Steroids

There is no evidence that steroids are of benefit in septic shock.

Box 6.3 Clinical features of septicaemia

History	Examination
Fever	Fever
Poor feeding	Tachycardia, tachypnoea, low blood pressure
Miserable, irritable, lethargy	Purpuric rash (meningococcal septicaemia) (Fig. 6.9).
History of focal infection, e.g. meningitis, osteomyelitis, gastroenteritis, cellulitis	Shock
Predisposing conditions, e.g. sickle cell disease, immunodeficiency	Multi-organ failure

Summary

Septicaemia

- The most common cause of septic shock in children is meningococcal disease
- May occur without meningitis
- Early antibiotic therapy and fluid resuscitation are life-saving
- May need admission to paediatric intensive care for multi-organ failure.

Coma

In coma, there is disturbance of the functioning of the cerebral hemispheres and/or the reticular activating system of the brainstem. The level of awareness may range from excessive drowsiness to unconsciousness.

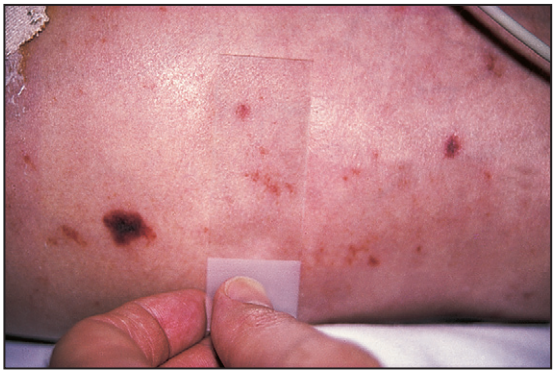


Figure 6.9 The glass test for meningococcal purpura. Parents are advised to suspect meningococcal disease if their child is febrile and has a rash that does not blanch when pressed under a glass. (Courtesy of Dr Parviz Habibi.)

Table 6.2 Glasgow Coma Scale, incorporating Children's Coma Scale

	Glasgow Coma Scale (4–15 years)	Children's Coma Scale (<4 years)	
	Response	**Response**	**Score**
Eyes	Open spontaneously	Open spontaneously	4
	Verbal command	React to speech	3
	Pain	React to pain	2
	No response	No response	1
Best motor response			
Verbal command	Obeys	Spontaneous or obeys verbal command	6
Painful stimulus	Localises pain	Localises pain	5
	Withdraws	Withdraws	4
	Abnormal flexion	Abnormal flexion (decorticate posture)	3
	Extension	Abnormal extension (decerebrate posture)	2
	No response	No response	1
Best verbal response	Oriented and converses	Smiles, orientated to sounds, follows objects, interacts	5
	Disoriented and converses	Fewer than usual words, spontaneous irritable cry	4
	Inappropriate words	Cries only to pain	3
	Incomprehensible sounds	Moans to pain	2
	No response	No response to pain	1

A score of <8 out of 15 means that the child's airway is at risk and will need to be maintained by a manoeuvre or adjunct.

It is assessed rapidly by using AVPU (**A**lert, responds to **V**oice, responds to **P**ain, **U**nresponsive) or the Glasgow Coma Scale (Table 6.2).

The immediate assessment of a child in coma is shown in Figures 6.10 and 6.11. The causes, clinical features and investigations of coma are listed in Table 6.3. In contrast to adults, most children have a diffuse metabolic insult rather than a structural lesion.

The history, examination and investigation of coma are directed towards the cause. Early treatment of treatable causes, especially hypoglycaemia and infection, is paramount. Raised intracranial pressure is treated with:

- the head positioned midline
- the head end of the bed tilted by 20–30°
- isotonic fluids at 60% maintenance
- intubation and ventilation if Glasgow Coma Score <9
- mannitol or 3% saline as osmotic diuretics
- maintaining normothermia and high normal blood pressure.

An intracranial mass lesion may require neurosurgical intervention.

Status epilepticus

This is a seizure lasting 30 minutes or longer, or when successive seizures occur so frequently that the patient does not recover consciousness between them. After immediate primary assessment and resuscitation, the priority is to stop the seizure as quickly as possible (Fig. 6.12).

Anaphylaxis

Anaphylaxis is rapid in onset and may be fatal. It has an incidence of one episode every 20 000 person years, and about 1 in 1000 cases are fatal. In children, 85% of anaphylaxis is caused by food allergy; most are IgE-mediated reactions with significant respiratory or cardiovascular compromise. Other causes include insect stings, drugs, latex, exercise, inhalant allergens and idiopathic. While most paediatric anaphylaxis occurs in children <5 years, when food allergy is most prevalent, the majority of fatal paediatric anaphylaxis occurs in adolescents with allergy to nuts; asthma is an additional risk factor. The acute management of anaphylaxis relies

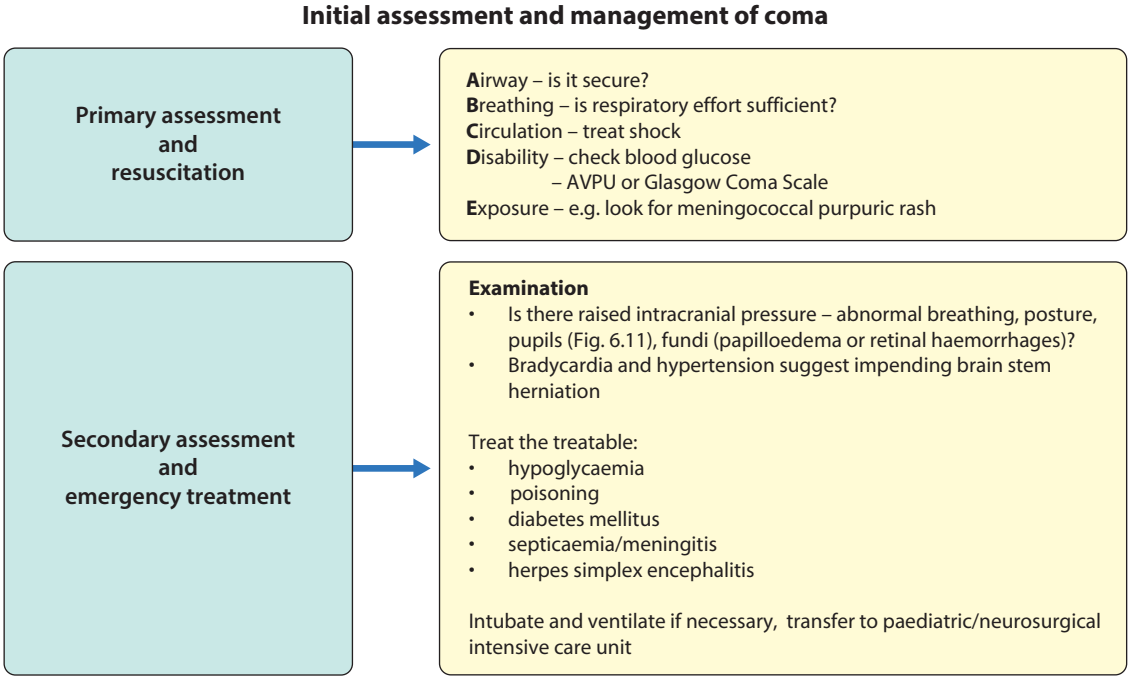


Figure 6.10 Initial assessment and management of coma.

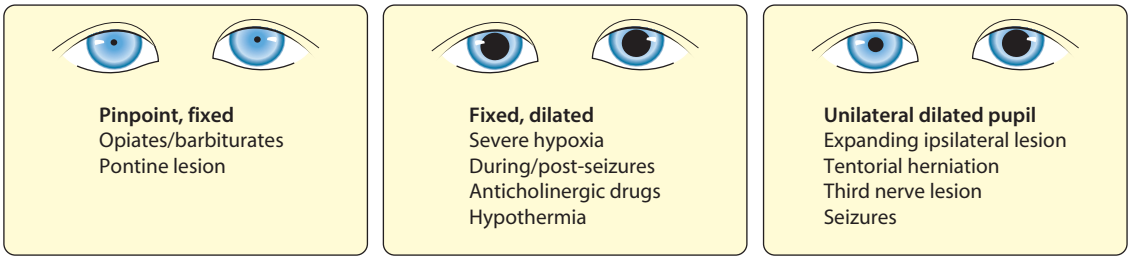


Figure 6.11 Pupillary signs in coma.

on early administration of adrenaline (epinephrine) (Fig. 6.13). Long-term management involves detailed strategies and training for allergen avoidance, a written management plan with instructions for the treatment of allergic reactions and the provision of adrenaline (epinephrine) auto-injector(s). In some cases, such as insect sting anaphylaxis, allergen immunotherapy may be effective in preventing future episodes. The experience of an anaphylactic reaction can have a significant psychological impact on the child and family.

Summary

Anaphylaxis in children/adolescents

- Reaction is mainly to foods – 1 in 1000 episodes is fatal
- Risk factors for fatal outcome include adolescent age group, coexistent asthma and nut allergy
- Acute management is early administration of intramuscular adrenaline (epinephrine).

Apparent life-threatening events (ALTE)

These occur in infants and are a combination of apnoea, colour change, alteration in muscle tone, choking or gagging, which are frightening to the observer. They are most common in infants less than 10 weeks old and may occur on multiple occasions. They may be the presentation of a potentially serious disorder, although often no cause is identified.

Management requires a detailed history and thorough examination to identify problems with the baby or in caregiving. The infant should be admitted to hospital. Causes and investigations to be considered are listed in Box 6.4. Multi-channel overnight monitoring is usually indicated.

In most, the episode is brief, with rapid recovery, and the baby is well clinically. Baseline investigations and overnight monitoring of oxygen saturation, respiration and ECG are found to be normal. The parents should be taught resuscitation and will find it helpful to receive follow-up from a specialist paediatric nurse and paediatrician.

Table 6.3 Causes, history and examination and investigation of coma

Cause	History and examination	Diagnostic investigations
Infection Meningitis or meningoencephalitis	Fever Irritability, lethargy, drowsiness Poor feeding, vomiting Rash, e.g. meningococcal purpura Seizures Neck stiffness and pain; bulging fontanelle Overseas travel	Full blood count Culture of blood, urine, infected sites, CSF (unless contraindicated) for bacteria and viruses Acute-phase reactant Rapid bacterial antigen/PCR tests for organisms
Status epilepticus or post-ictal	Past history of seizures Neurocutaneous lesions on the skin Developmental delay Ongoing seizure activity, e.g. abnormal eye movements Focal neurological signs	Blood glucose Electrolytes – sodium, potassium, calcium, magnesium Drug levels if on anticonvulsants EEG CT scan
Trauma – accidental/ non-accidental	History of road traffic accident, fall, etc. Bruising, haemorrhage Fractures – cervical spine, etc. Focal neurology Retinal haemorrhages	Radiological – plain X-rays or CT/MRI scans
Intracranial tumour or haemorrhage/ infarct/abscess	Symptoms or signs of raised intracranial pressure Focal neurological signs, e.g. squint	Cranial CT/MRI scan Haemorrhage – coagulation screen, screen for procoagulant disorders (protein C/S deficiency)
Metabolic		
1. Diabetes mellitus	Previously diagnosed diabetes mellitus Diabetic ketoacidosis	Blood glucose, plasma electrolytes Urine for glucose and ketones Blood gas analysis
2. Hypoglycaemia	Any acutely ill child Known diabetes mellitus Sudden onset of coma	Low blood glucose
3. Inborn errors of metabolism	Previous history of loss of consciousness Sudden collapse Consanguinity, death or illness of siblings Developmental delay Hepatomegaly	Blood glucose Blood gas analysis Blood ammonia, lactate Urine amino and organic acids Plasma amino acids
4. Hepatic failure	Jaundice Abnormal bleeding	Abnormal liver function tests Prolonged prothrombin time
5. Acute renal failure	Oliguria, hypertension	Abnormal creatinine
Poisoning	Accidental – poison usually identified Deliberate – tablets may be found, also illicit drugs and alcohol	Toxicology screen Plasma level for paracetamol and salicylates
Shock	Septicaemia Dehydration Cardiac failure	Full blood count and cultures Urea, electrolytes, blood gas
Hypertension	Symptoms and signs of raised intracranial pressure Fundoscopy – hypertensive changes	Left ventricular hypertrophy on ECG or echocardiography Creatinine and electrolytes
Respiratory failure	Respiratory failure	Chest X-ray Arterial blood gas – hypoxia, hypercarbia

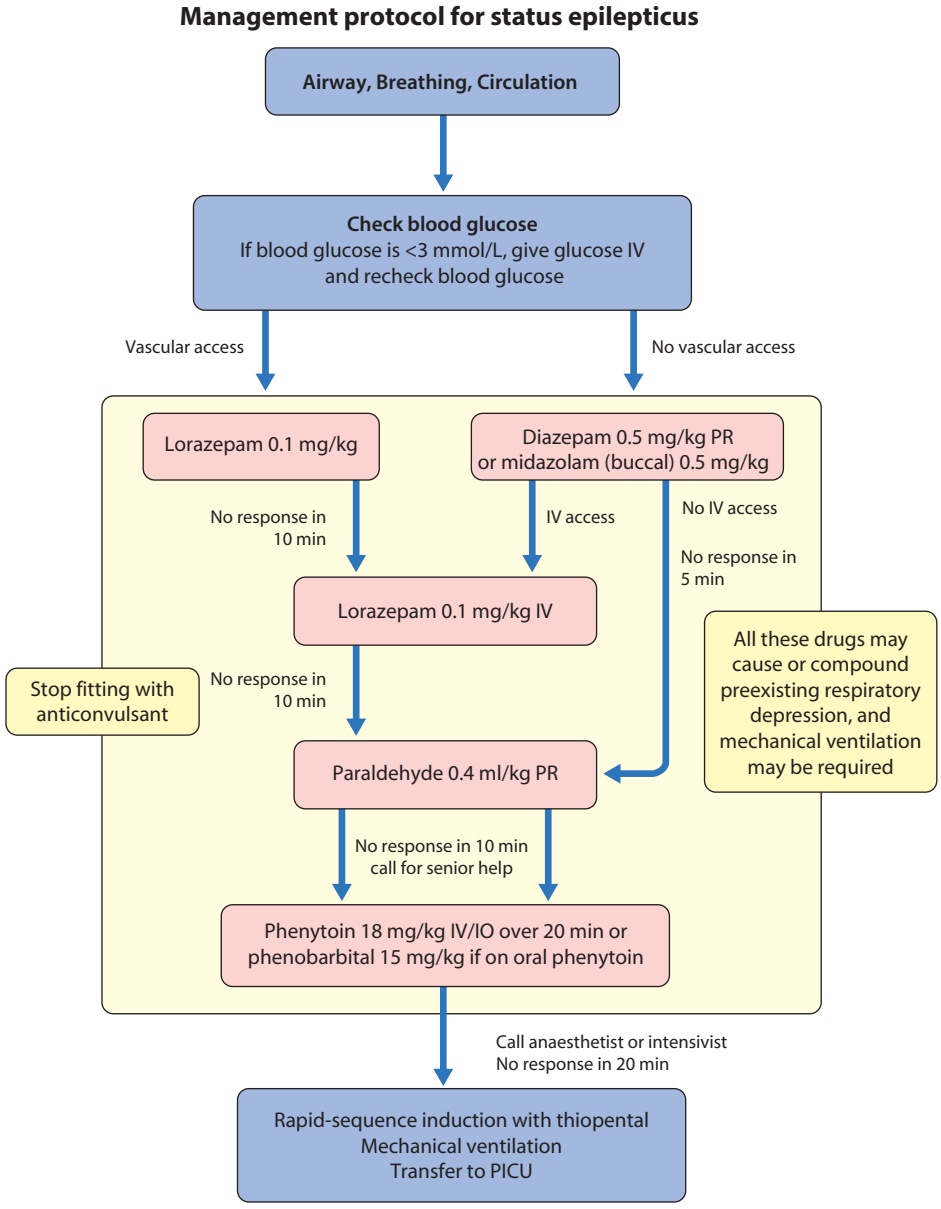


Figure 6.12 Management protocol for status epilepticus. (Adapted from Advanced Life Support Group. 2005. *Advanced Paediatric Life Support. The Practical Approach*, 4th edn. Blackwell BMJ Books, London.)

Detailed specialist investigation and assessment will be required if clinical, biochemical or physiological abnormalities are identified.

The death of a child

The risk of death is four times greater during infancy than at any other age in childhood. In many, a serious condition will have been diagnosed before or after birth, such as a congenital abnormality or complications of prematurity. Deaths which occur suddenly and unexpectedly in infancy are known as sudden unexpected death in infancy (SUDI). In some, a previously undiagnosed congenital abnormality, e.g. congenital heart disease, will be found at autopsy. Rarely, an inherited metabolic disorder is identified, in particular the fatty acid oxidation defect medium-chain acyl-CoA dehydrogenase deficiency (MCAD), which can very rarely result in sudden death in infants, but is increasingly identified in the UK from routine biochemical screening (Guthrie test). After 1 month of age, in most instances of sudden death in a previously well infant, no cause is identified and the death is classified as sudden infant death syndrome (SIDS). The vast majority of such deaths, even when occurring more than once in the same family, are due to natural causes. Rarely, the death may be due to suffocation or other forms of non-accidental injury.

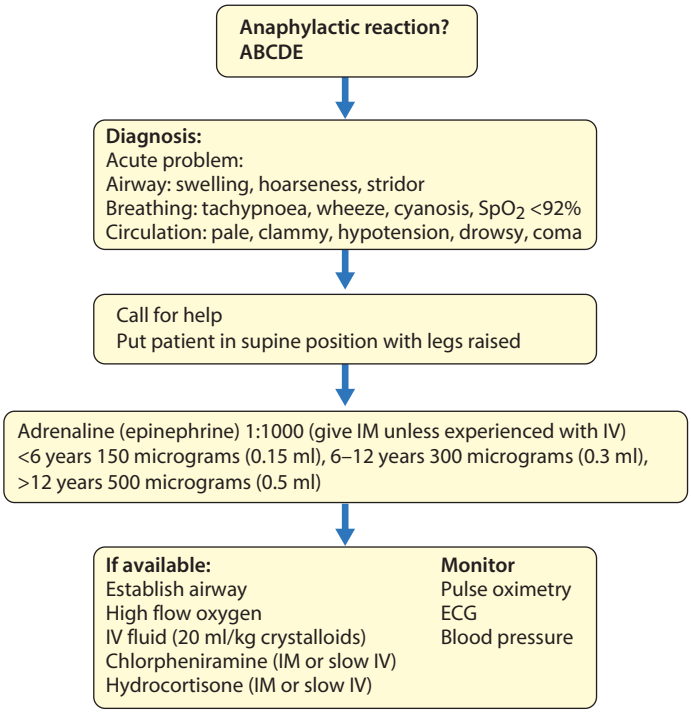


Figure 6.13 Emergency treatment of anaphylaxis. (Adapted from Resuscitation Council (UK) 2008 Resuscitation Guidelines.)

Box 6.4 Causes and investigations to be considered in apparent life-threatening events

Causes

- Infections – respiratory syncytial virus (RSV), pertussis
- Seizures
- Gastro-oesophageal reflux (present in one-third of normal infants)
- Upper airways obstruction – natural or imposed
- No cause identified

Uncommon causes

- Cardiac arrhythmia
- Breath-holding
- Anaemia
- Heavy wrapping/heat stress
- Central hypoventilation syndrome
- Cyanotic spells from intrapulmonary shunting

Investigations to be considered

- Blood glucose (as soon as possible)
- Blood gas (as soon as possible)
- Oxygen saturation monitoring
- Cardiorespiratory monitoring
- EEG
- Oesophageal pH monitoring
- Barium swallow
- Full blood count
- Urea and electrolytes, liver function tests
- Lactate
- Urine (collect and freeze first sample)
  - metabolic studies
  - microscopy and culture
  - toxicology
- ECG – for QT_c conduction pathway abnormality
- Chest X-ray
- Lumbar puncture.

Sudden infant death syndrome

This is defined as the sudden and unexpected death of an infant or young child for which no adequate cause is found after a thorough postmortem examination. There is marked variation in the incidence of SIDS in different countries, suggesting that environmental factors are important (Box 6.5). SIDS occurs most commonly at 2–4 months of age (Fig. 6.14). The risk for subsequent children is slightly increased.

In the UK, the incidence of SIDS has fallen dramatically during the last 20 years (Fig. 6.15), coinciding with a national 'Back to Sleep' campaign (Fig. 6.16). This advocates that:

- Infants should be put to sleep on their back (not their front or side)
- Overheating by heavy wrapping and high room temperature should be avoided
- Infants should be placed in the 'feet to foot' position

Box 6.5 Factors associated with SIDS

The infant

- Age 1–6 months, peak at 12 weeks
- Low birthweight and preterm (but 60% are normal birthweight term infants)
- Sex (boys 60%)
- Multiple births

The parents

- Low income[a]
- Poor or overcrowded housing
- Maternal age (mother aged <20 years has three times the risk of a mother aged 25–29 years, but 80% of affected mothers are >20 years old)[a]
- Single unsupported mother (twice the rate of supported mothers)
- High maternal parity[a]
- Maternal smoking during pregnancy (1–9 cigarettes/day doubles the risk: >20/day increases the risk five-fold)[a]
- Parental smoking after baby's birth

The environment

- The infant sleeps lying prone
- The infant is overheated from high room temperature and too may clothes and covers, particularly when ill.

[a]Three of these four factors are present in over 40% of SIDS but only 8% of control families.
Based on data from Fleming P, Blair P, Bacon C et al. 2000. *Sudden Unexpected Deaths in Infancy*. The Stationery Office, London.

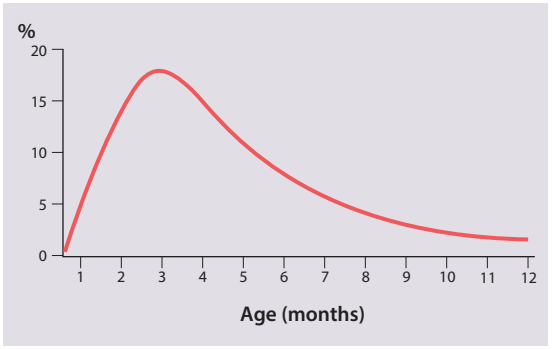


Figure 6.14 Age distribution of SIDS. (Based on data from Fleming P, Blair P, Bacon C et al. 2000. *Sudden Unexpected Deaths in Infancy*. The Stationery Office, London, with permission.)

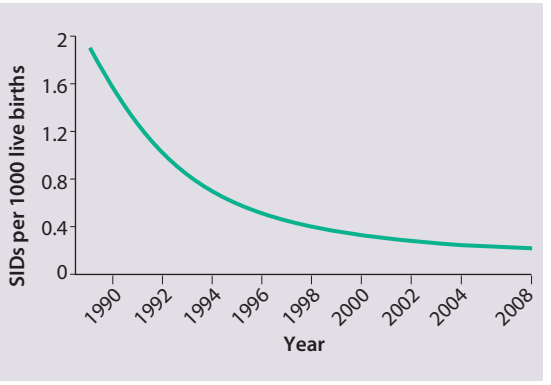


Figure 6.15 Decline in the number of deaths from SIDS in the UK from 1.9/1000 live births in 1989 to 0.31 in 2008.

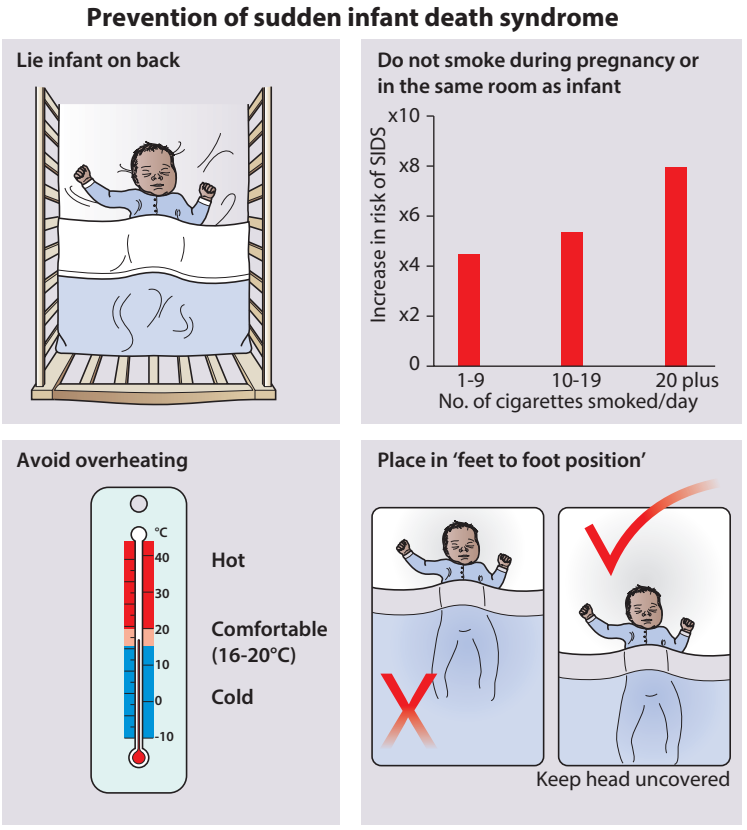


Figure 6.16 Key features of the 'Back to Sleep' campaign.

Management of the sudden unexpected death of an infant

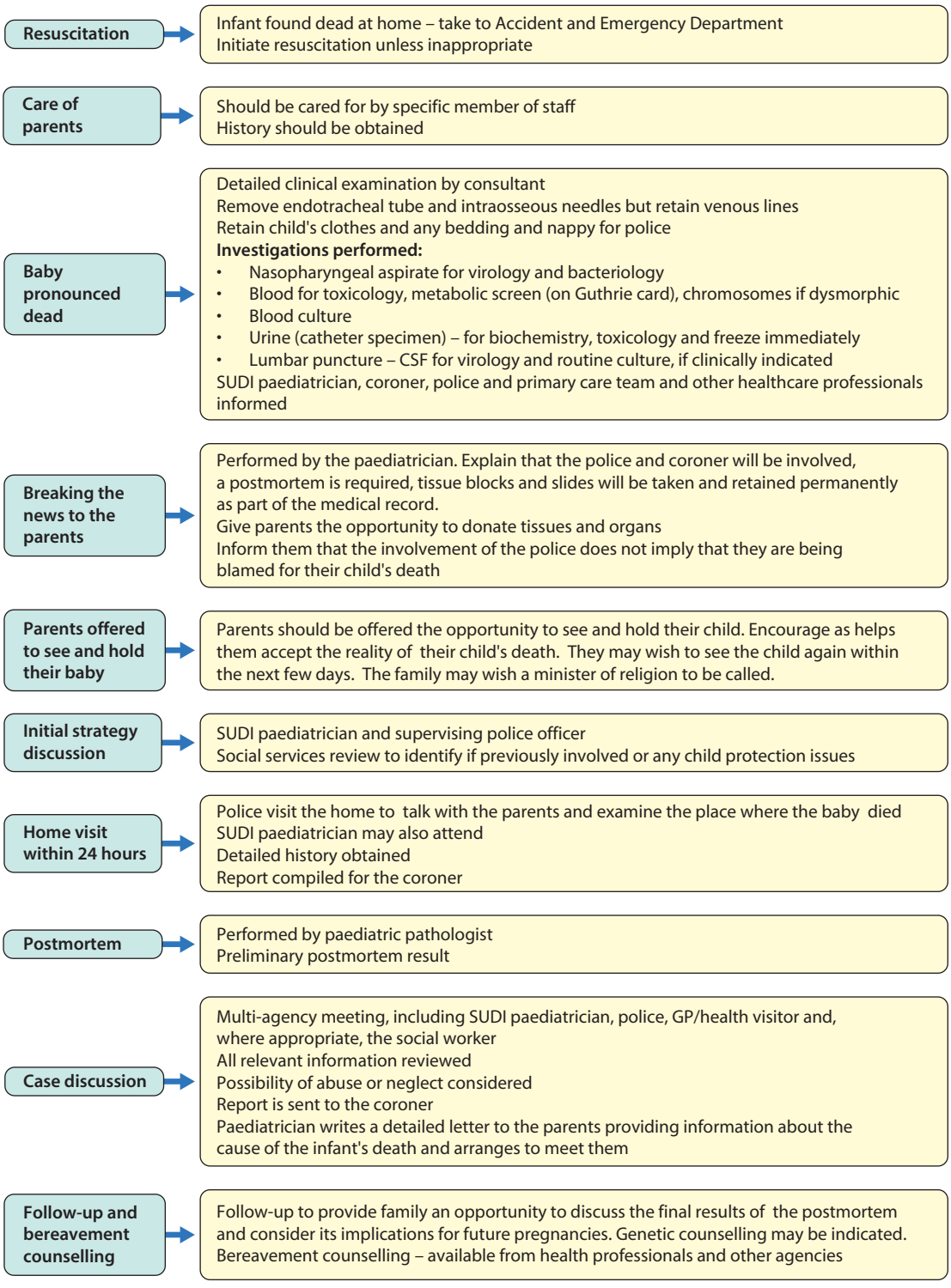


Figure 6.17 A recommended approach to the management of the sudden unexpected death of an infant. There are local variations in its implementation. (Adapted from RCPCH. 2004. *Sudden Unexpected Death in Infancy*. The report of a working group convened by the Royal College of Pathologists and The Royal College of Paediatrics and Child Health. RCPCH, London. Available at: www.rcpath.org and www.rcpch.ac.uk (Accessed January 2011).

- Parents should not smoke near their infants
- Parents should seek medical advice promptly if their infant becomes unwell.
- Parents should have the baby in their bedroom for the first 6 months of life
- Parents should avoid bringing the baby into their bed when they are tired or have taken alcohol, sedative medicines or drugs
- Parents should avoid sleeping with their infant on a sofa, settee or armchair.

Summary

Sudden infant death syndrome (SIDS)

- SIDS is the commonest cause of death in children aged 1 month to 1 year.
- The peak age is 2–4 months.
- SIDS has been dramatically reduced by lying babies on their back to sleep.

Following the sudden death of a child

The sudden death of a child is one of the most distressing events that can happen to a family. If close family members are absent, arrangements should be made for them to come, if this is possible. The family should be spoken to sympathetically and in private (see Ch. 5). An outline of the recommended management after an infant has died suddenly and unexpectedly is shown in Figure 6.17.

Summary

Following an unexpected death

- The parents should be offered the opportunity to see and hold their child.
- The coroner must be informed and a postmortem performed.
- The parents should be informed about the postmortem, that investigations will be performed and that the police will be involved and they should be reassured that this is standard practice.
- Follow-up and bereavement counselling should be offered.

Further reading

Advanced Life Support Group: *Advanced Paediatric Life Support. The Practical Approach*, ed 4, Blackwell BMJ Books, 2005, London.

Goldman A, Hain R, Lieben S: *The Oxford Textbook of Palliative Care in Children*, ed 2, Oxford, 2011, Oxford University Press.

Medical, psychological and practical issues of caring for terminally ill children and their families.

RCPCH: *Sudden Unexpected Death in Infancy*. The report of a working group convened by the Royal College of Pathologists and The Royal College of Paediatrics and Child Health, London, 2004, RCPCH. Available at: www.rcpath.org and www.rcpch.ac.uk (Accessed May 2011).

Resuscitation Council (UK): Updated guidelines on paediatric life support, London, 2008, 2010. Resuscitation Council (UK).

7

Accidents, poisoning and child protection

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Children need a safe, healthy and nurturing environment to achieve their full potential. Environmental hazards include accidents and poisons. Protecting children and young people from harm is a primary responsibility of parents, families and carers, but all members of society, including doctors, teachers and other professionals, play an important role in advocating for safe environments for children.

The risk of accidents, poisons and abuse is increased by:

- Poverty
- Poor-quality, overcrowded homes
- Lack of a safe environment for play
- Poor parenting skills, which may be due to parental psychiatric illness, drug and alcohol abuse, violent temperament, poor education or lack of social support.

Accidents

Accidents (now often called 'unintentional injuries') are extremely common. In the UK, one in four children attend an A&E department each year; about half because of an accident. They are more common in boys. Most accidents cause only minor injury but can also be fatal. Injuries, mostly accidents, and poisoning are the major cause of death in children 1–14 years of age in the UK (Figs 7.1, 7.2). Accidents also cause significant disability and suffering, including post-traumatic stress disorder. Head injury with brain damage is the major cause of disability from accidents. Cosmetic damage following burns, scalds and other accidents may cause the child profound psychological harm.

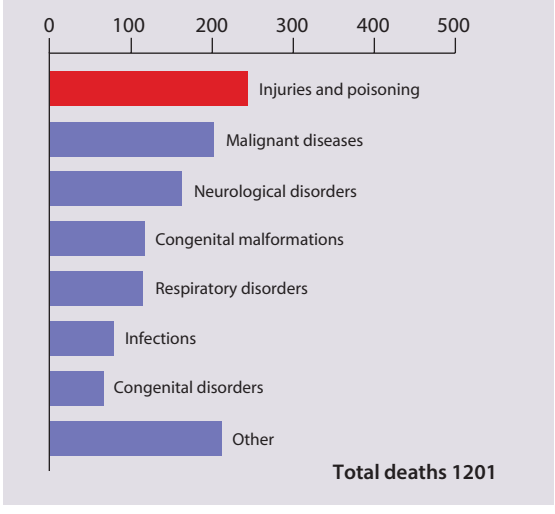


Figure 7.1 Causes of death in children aged 1–14 years in England and Wales (2008).

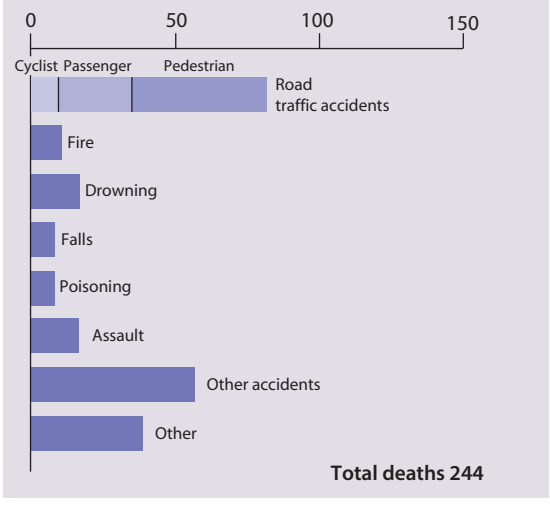


Figure 7.2 Causes of fatal injuries and poisoning in children in England and Wales (2008). The most common cause of injuries is road traffic accidents.

Injuries are the most common cause of death in children over 1 year of age.

Types of accident affecting children

Most accidents affecting children depend on the child's age and development.

In babies and toddlers they can usually be anticipated by an observant adult and may be prevented by vigilant supervision.

Accident prevention

The prevention of childhood accidents is clearly important. Doctors and nurses who treat children and see the effects of accidents are particularly well placed to provide the community with advice on appropriate preventive measures (Fig. 7.3). In order to prevent accidents:

- The relationship between an individual type of accident and the child's developmental level must be considered
- Specific solutions should be based on detailed epidemiology, e.g. child-resistant containers are the best method to limit children's access to medicines
- Changes backed by legislation are the most successful.

Organisations such as the Royal Society for the Prevention of Accidents and the Children Accident Prevention Trust are important resources in helping reduce accidents by providing education and lobbying for legislative changes, e.g. bicycle helmet wearing.

The number of children killed in accidents has declined markedly over the past 20 years.

Road traffic accidents

Road traffic accidents (RTAs) are the most common cause of accidental death or serious injury in childhood and in 2008 accounted for 23 000 children killed or injured on the UK's roads. RTAs can be divided into several types.

Pedestrian road traffic accidents

Children's involvement in road traffic accidents is mostly as pedestrians. Boys between the ages of 5 and 9 years are at maximum risk, particularly after school. Children are unable to estimate the speed or dangers of traffic and to foresee dangerous situations. Social and economic factors play a part, with children in social class 5 being 5 times more likely to be killed as a pedestrian than children in social class 1.

Child passengers in cars

Unrestrained children become missiles inside cars during crashes, even at low speeds. There is good evidence that child restraint systems prevent injury and

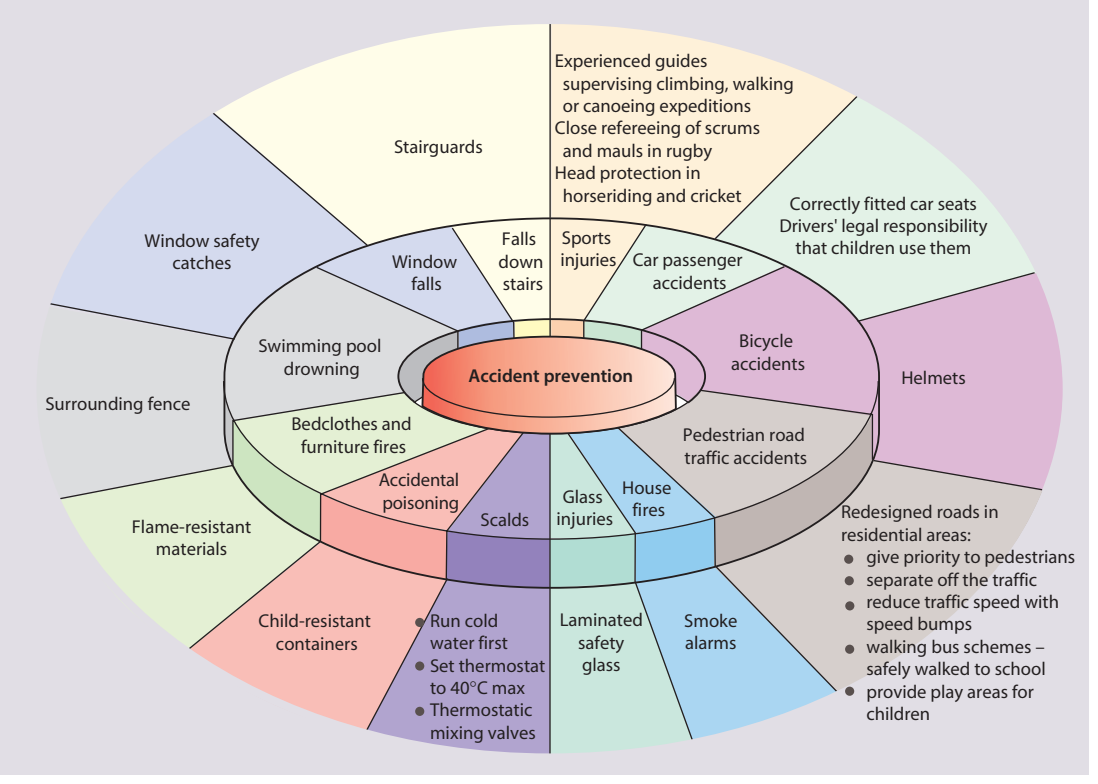


Figure 7.3 Examples of accident prevention.

death and they are required by law in the UK for all children <135 cm tall.

Bicycle accidents

Bicycle accidents are common during childhood. A boy has a 1 in 80 chance of having a cycling-related head injury severe enough to warrant admission to hospital during childhood. Although 70% of cycle deaths involve a head injury, only 1 in 5 children >10 years wear a helmet.

Head injuries

Minor head injuries in childhood are common, and the vast majority of children recover without suffering any ill effects. However, about 1 in 800 of these children develop serious problems. The aim of the management of head injuries is to identify those children requiring treatment and to avoid secondary damage to the brain from hypoxia or poor cerebral perfusion (Fig. 7.4).

Head injury may result in concussion, a reversible impairment of consciousness, a subdural or extradural haematoma or intracerebral contusion (see Ch. 27).

In infants, as their skull sutures have not fused, cranial volume may increase from an extradural or subdural bleed before neurological signs or symptoms develop. The haemoglobin concentration may fall and they may become shocked.

Summary

Accidental head injury management

- Mild – discharge home with written advice
- Potentially severe – monitor to avoid secondary damage
- Severe – resuscitate, CT scan and neurosurgical referral.

Internal injuries

Children may suffer internal injuries associated with severe trauma. These include:

- Abdominal injuries, including a ruptured spleen, ruptured liver, kidney and bowel. There should be a high index of suspicion for these internal injuries if there has been abdominal trauma or if the clinical setting suggests significant inflicted, i.e. abusive, injury. The child needs close observation. Abdominal ultrasound (focused abdominal sonography in trauma – FAST scan) and X-rays, including CT scan, may be helpful. If there is any doubt, a laparotomy/laparoscopy is undertaken. Intra-abdominal injuries such as a contained splenic bleed are increasingly managed conservatively with close monitoring, but paediatric surgical support must be available immediately in case surgery is required.
- Chest injuries, including pneumothorax and haemopericardium, may require emergency treatment. These children should be managed in a paediatric intensive care unit.

Burns and scalds

Burns and scalds are a significant accidental (and non-accidental) cause of death, although most of the deaths occurring in house fires are caused by gas and smoke inhalation rather than thermal injury. Scalds in toddlers are common; children are scalded at lower temperatures than adults, as their skin is thinner. It is important to exclude inflicted burns.

Management

The severity of the injury is assessed:

- Is the *airway, breathing and circulation* satisfactory?
- *Was there any smoke inhalation*? If this has occurred, there is a danger of subsequent respiratory complications and carbon monoxide poisoning. All affected children should be observed and managed in hospital, with a low threshold to protect the airway before secondary problems develop.
- *Depth of the burn*. In superficial burns, the skin will be epithelialised from surviving cells. In partial thickness burns, there is some damage to the dermis with blistering, and the skin is pink or mottled; regeneration for superficial and partial thickness burns is from the margins of the wound and from the residual epithelial layer surrounding the hair follicles deep within the dermis. In deep (full thickness) burns, the skin is destroyed down to and including the dermis and looks white or charred, is painless and involves hair follicles, hence skin grafting is often required. Deep burns need assessment and treatment in hospital.
- *Surface area of the burn*. This should be calculated from a surface area chart (Fig. 7.5). The palm and adducted fingers cover about 1% of the body surface. Burns covering more than 5% full thickness and 10% partial thickness need assessment by burns specialists. Involvement of more than 70% of the body surface carries a poor chance of survival.
- *Involvement of special sites*. Burns to the face may be disfiguring, those to the mouth may compromise the airway from oedema, and those to the hand/joints may cause functional loss from scarring. Burns to the perineum and other special sites should all be referred to a burns unit.

Treatment

This should be directed at:

- relieving pain, assessed with a pain score; may require the use of strong analgesics such as intravenous morphine
- treating shock with intravenous fluids, preferably plasma expanders, and close monitoring of haematocrit and urinary output. Children with more than 10% burns will require intravenous fluids
- providing wound care. Burns should be covered with cling film (plastic wrapping), which reduces pain from contact with cold air and reduces the risk of infection. Blisters should be left alone. Irrigation

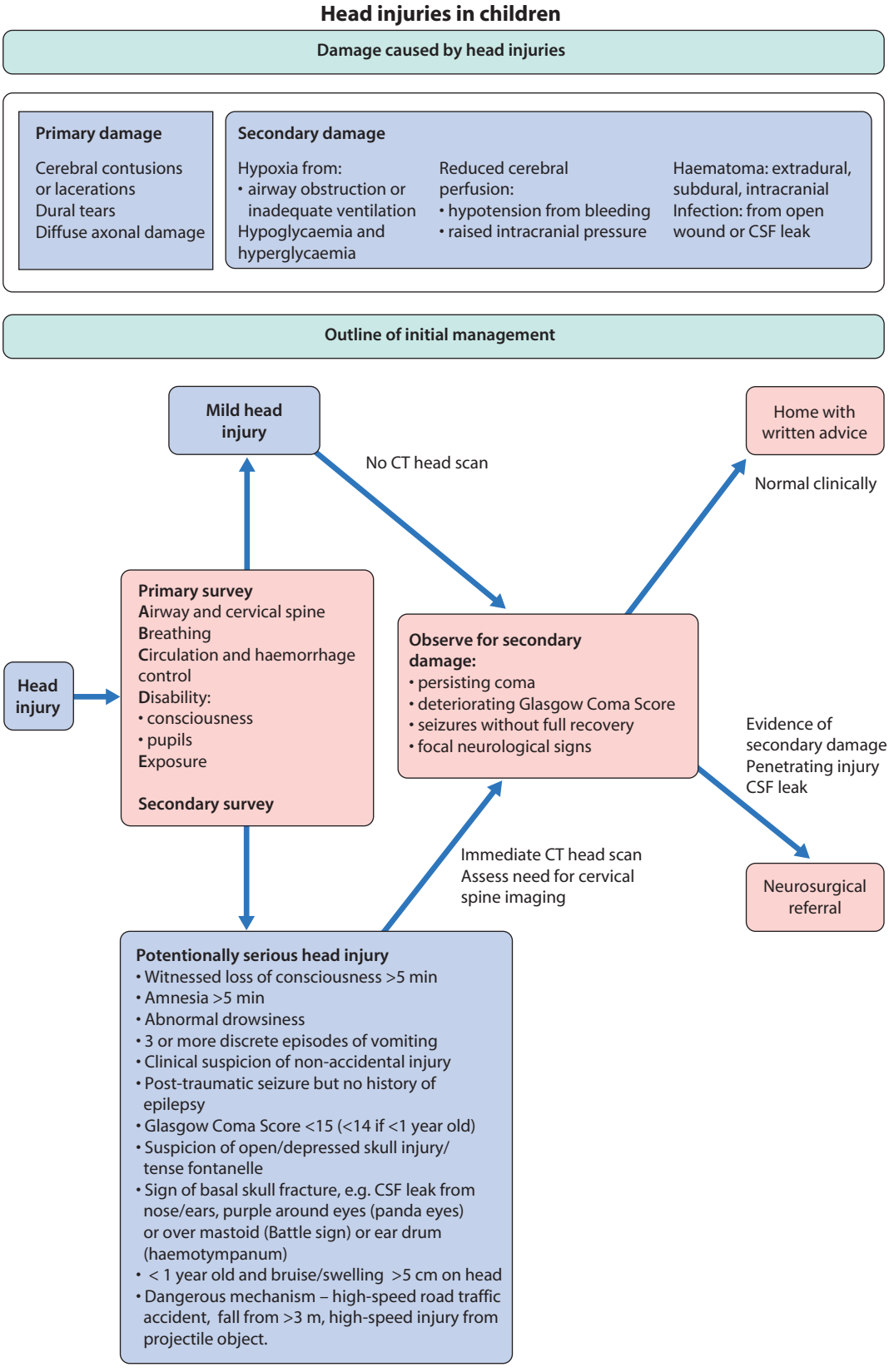
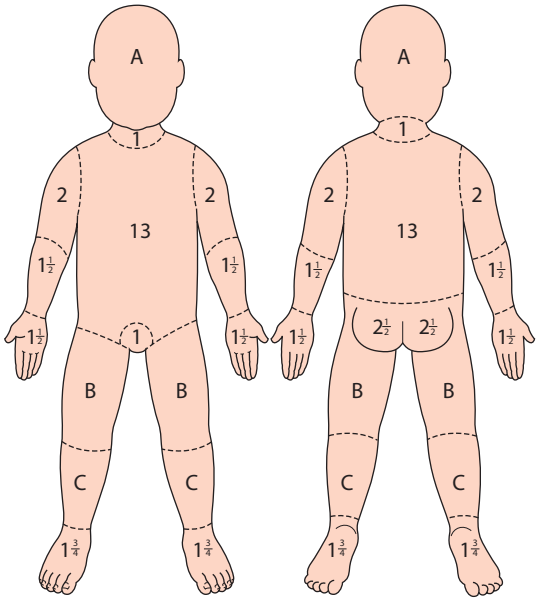


Figure 7.4 Head injuries in children. (Adapted from NICE Guideline, 2007.)



Area indicated	Surface area at			
	1 year	5 years	10 years	15 years
A	8.5	6.5	5.5	4.5
B	3.25	4.0	4.5	4.5
C	2.5	2.75	3.0	3.25

Figure 7.5 Method of calculating the surface area of a burn (Lund and Browder chart).

with cold water should only be used briefly to superficial or partial thickness burns covering less than 10% of the body as it may rapidly cause excessive cooling. Tetanus immunisation status must be ascertained and a booster given if required. Ongoing care involves removal of dead tissue and placement of sterile dressings.

Severe burns or significant burns to special sites are best dealt with in specialist units. Plastic surgeons will often need to embark on a programme of skin grafts and treatment of contractures. The psychological sequelae of severe burns are often marked and long-lasting, and appropriate psychological support is required.

Summary

Burns and scalds

- Assess: need for resuscitation, if any smoke inhalation, depth and surface area of burns and if special sites involved
- Management: pain control, treat shock, provide wound care (cling film), refer to specialist centre if more than 5% full thickness, 10% partial thickness or significant burns to special sites.

Drowning and near-drowning

Drowning is a significant cause of accidental death in children in the UK. Most victims are young children. Drowning is three times more common in boys than in girls. Warmer, affluent countries tend to have a higher incidence of drowning than in the UK, particularly because of drowning in domestic swimming pools. Babies may drown in the bath, toddlers may wander into domestic ponds or swimming pools, and older children may get into difficulty in swimming pools, rivers, canals, lakes and in the sea. Children should always be supervised when swimming.

Near-drowning

Up to 30% of fatalities can be prevented by skilled on-site resuscitation. Even children who are unconscious with fixed dilated pupils can survive near-drowning episodes, particularly if the water is cold, due to the protective effect of hypothermia against hypoxic brain injury. Children who are unconscious with fixed dilated pupils should therefore be fully resuscitated until their temperature is nearly normal. Immediate management at the waterside is with mouth-to-mouth resuscitation and chest compressions. Heat loss should be prevented by covering and warming. Children who may have inhaled water should be admitted to hospital to be observed for signs of respiratory distress from pulmonary oedema after 1–72 h from secondary surfactant. Some aspirate water and develop pneumonia with secondary infection. It is now thought that there is no difference in outlook for fresh- and salt-water drowning.

Choking, suffocation and strangulation

Children may choke on vomit, toys or food. Some children may strangle themselves accidentally on curtain cords, bedding and necklaces. Most are accidents but some such injuries are inflicted deliberately as a form of child abuse. Some adolescents deliberately hang themselves.

In airway obstruction from an aspirated foreign body, the actions outlined in Figures 7.6 and 7.7 should be followed.

Dog bites

Most dog bites are minor, but severe lacerations, particularly to the face, do occur, especially in the toddler age group.

Wound management is as important as antimicrobials in preventing infection:

- Copious wound irrigation, toilet and debridement
- Removal of foreign bodies
- Delayed wound closure, especially on limbs, due to increased incidence of infection
- Raise and immobilise limb
- Regular wound review
- Tetanus booster
- Prophylactic antibiotics if wounds or child is at increased infection risk, i.e. puncture wounds, bites to hands/wrists, crush wounds with devitalised

Inhaled foreign body

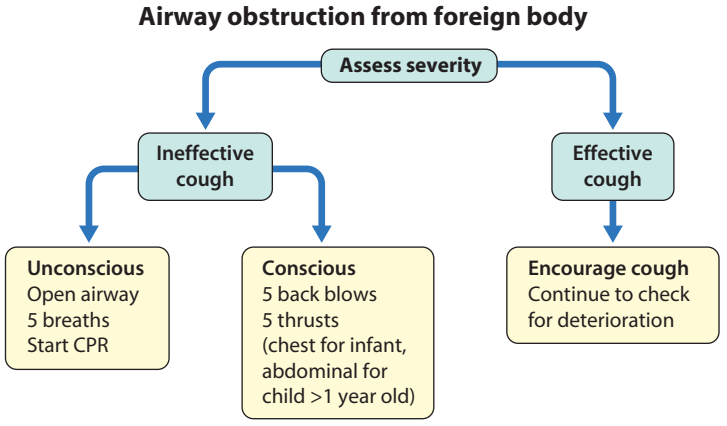


Figure 7.6 Management of a foreign body.



Figure 7.7a Abdominal thrusts using the Heimlich manoeuvre in older children to expel an inhaled foreign body. One hand is formed into a fist and placed against the child's abdomen above the umbilicus and below the xiphisternum. The other hand is placed over the fist. Both hands are thrust into the abdomen. This is repeated several times. The child can be standing, kneeling, sitting or supine.

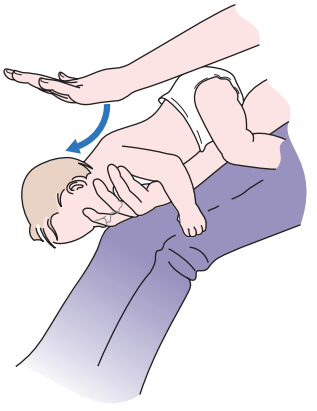


Figure 7.7b In infants, back blows and chest thrusts are recommended to expel an inhaled foreign body. Abdominal thrusts are best avoided in infants as they may cause intra-abdominal injury.

tissue, bites to genitals, child has diabetes mellitus, immunosuppression and in all bites after primary closure.

The antibiotic of choice is co-amoxiclav, as this also covers *Pasteurella* infection.

Although there has been much publicity about fierce dog breeds, such as Rottweilers, attacking children in parks or public places, most attacks are by dogs known to the child.

Poisoning

Poisoning in children may be:

- accidental – the vast majority
- deliberate self-poisoning in older children
- non-accidental as a form of child abuse
- iatrogenic.

Accidental poisoning

Although many thousands of young children are rushed to doctors' surgeries or hospital for urgent medical attention following accidental ingestion, most do not develop serious symptoms, as they ingest only a small quantity of poison or take relatively non-toxic substances (Table 7.1). However, a small percentage of children become seriously ill and a very few children die from poisoning each year.

Most accidental poisoning is in young children, with a peak age of 30 months. Inquisitive toddlers are unaware of the potential danger of taking medicines,

Table 7.1 Potential toxicity in accidental poisoning in infants and young children, with some examples

Toxicity	Medicines	Household products	Plants
Low	Oral contraceptives, most antibiotics	Chalk and crayons, washing powder	Cyclamen, sweet pea
Intermediate	Paracetamol elixir, salbutamol	Bleach, disinfectants, window cleaners	Fuchsia, holly
High	Alcoholic drinks, digoxin, iron, salicylate, tricyclic antidepressants	Acids, alkalis, petroleum distillates, organophosphorus insecticides	Deadly nightshade, laburnum, yew

household products and eating plants. Most ingestions occur in the child's own home, when supervision is inadequate. Supervision of toddlers entails not only reacting to a dangerous situation but also prevention through anticipation.

The aim of management of poisoning should be to prevent unnecessary admissions to hospital while maintaining safety. There has been a marked reduction in the hospital admission rate for poisoning. Reasons for this include:

- The introduction of child-resistant containers – in the UK they must be used for paracetamol and salicylate preparations, and certain household products, such as white spirit; an alternative container for tablets is opaque blister packs
- The reduction in the number of tablets per pack
- A reduction in prescriptions of potentially harmful medicines, e.g. aspirin and iron.

Management

Clinical features that may alert one to poisoning are shown in Table 7.2; however, in young children parents usually know the identity of tablets or other poisons taken (Table 7.3). Management is outlined in Figure 7.8.

Table 7.2 Clinical features of poisons

Tachypnoea	Aspirin, carbon monoxide
Slow respiratory rate	Opiates, alcohol
Hypertension	Amphetamines, cocaine
Hypotension	Tricyclics, opiates, β-blockers, iron (secondary to shock)
Convulsions	Tricyclics, organophosphates
Tachycardia	Cocaine, antidepressants, amphetamines
Bradycardia	β-blockers
Large pupils	Tricyclics, cocaine, cannabis, amphetamines
Small pupils	Opiates, organophosphates

Summary

Accidental poisoning in children

- Common in toddlers and young children
- Most do not cause serious illness
- When an ingestion has occurred, identify the agent and assess its toxicity to plan management
- Potentially harmful poisons in children are alcohol, acids and alkalis, bleach, digoxin, batteries, iron, paracetamol, petroleum distillates, salicylates and tricyclic antidepressants
- Assess the social circumstances behind why it happened.

Deliberate poisoning in older children

Older children are more likely to take significant amounts of poison than younger children. Substances that can be regarded as having intermediate toxicity when taken accidentally should be regarded as potentially toxic when taken deliberately. Poisoning in older children should be recognised as a serious symptom and an indication of child and family disturbance, so all children who take poisons deliberately should ideally be assessed by a child and adolescent psychiatrist and a social worker.

Chronic poisoning

Children can be poisoned by chronic exposure to chemicals and pollutants. An example from the past is mercury poisoning from teething powders, which used to cause 'pink disease', so-called because it resulted in red painful extremities. It also caused anorexia, weight loss and hypotonia. Now the commonest causes are lead ingestion and smoking.

Lead poisoning

Environmental lead levels are now much reduced. In the past, certain paints contained lead. Children are liable to be poisoned from chewing such paintwork or from inhalation when the paint is removed. This is still a problem in parts of the USA. Lead fumes from burning

Table 7.3 Potentially harmful poisons

Poison	Adverse effects	Management
Alcohol (accidental or experimenting by older children)	Hypoglycaemia Coma Respiratory failure	Monitor blood glucose. Intravenous glucose if necessary. Blood alcohol levels for severity
Acids and alkalis	Inflammation and ulceration of upper gastrointestinal tract leading to stenosis	No emesis/gastric lavage No chemical antidotes as they produce heat Early endoscopy
Digoxin	Arrhythmias, hyperkalaemia	Activated charcoal if <1 h after ingestion ECG monitoring, serum digoxin concentration
Disc or button batteries	Mild gastrointestinal symptoms Oesophageal stricture with large batteries (>20 mm) Corrosion of gut wall and perforation Break open with release of mercury – rare	Monitor progress with chest and abdominal X-rays – almost all passed within 2 days, no symptoms Remove batteries if in oesophagus or signs of disintegration. Some authorities recommend removal if not passed within 48 h to avoid danger of disintegration
Iron	*Initial*: vomiting, diarrhoea, haematemesis, melaena, acute gastric ulceration *Latent period of improvement* *Hours later*: drowsiness, coma, shock, liver failure with hypoglycaemia and convulsions *Long term*: gastric strictures	Serious toxicity if >60 mg/kg elemental iron Abdominal X-ray to count the number of tablets Serum iron levels Gastric lavage considered in severe cases if <1 h after ingestion Intravenous desferrioxamine for chelation
Paracetamol – large ingestion uncommon in young children as tablets are difficult to swallow and elixir is too sweet	Gastric irritation Liver failure after 3–5 days	Check plasma concentration after 4 h after ingestion. If >150 mg/kg paracetamol is thought to have been taken, or the plasma concentration is high, start intravenous acetylcysteine Monitor prothrombin time, liver function tests and plasma creatinine
Petroleum distillates (paraffin/kerosene, white spirit)	Aspiration causing pneumonitis	Emesis and gastric lavage contraindicated Usually no treatment required
Salicylates	Tinnitus, deafness, nausea, vomiting, dehydration Hyperventilation causing respiratory alkalosis. Later, metabolic acidosis. Hypoglycaemia. Disorientation	Measure plasma salicylate concentration Gastric lavage if <1 h. Give activated charcoal Monitor fluid and electrolyte balance Correct dehydration, electrolyte imbalance and acidosis. Dialysis
Tricyclic antidepressants	Sinus tachycardia Conduction disorders Dry mouth, blurred vision Agitation, confusion, convulsions, coma Hypotension Respiratory depression	Activated charcoal if within 1 h Cardiac monitoring. Treat arrhythmias conservatively with sodium bicarbonate Correct metabolic acidosis Treat convulsions with diazepam

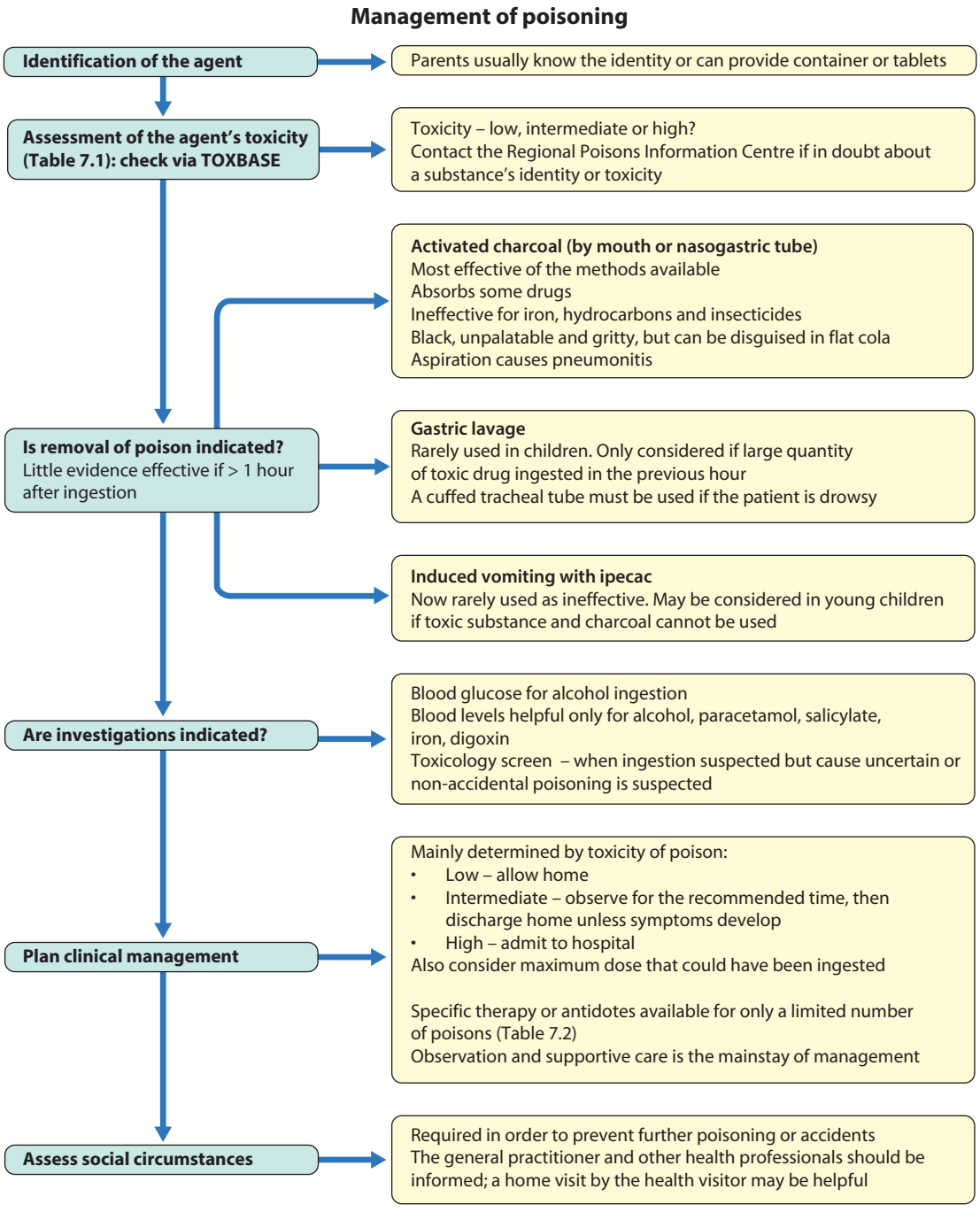


Figure 7.8 Outline of the management of poisoning. TOXBASE is an online database of the National Poisons Information Service for registered users.

batteries, lead shot for fishing and lead from old water pipes are other potential sources. Children from the Indian subcontinent may be poisoned by *surma*, the lead-containing eye make-up sometimes used even on young babies. Lead from vehicle exhaust fumes results in higher blood levels in children living in urban compared with rural areas. The change to unleaded petrol was in response to concern about its potential as an environmental hazard.

Children present with pica (compulsive eating of substances other than food), anorexia, colicky abdominal pain, irritability and failure to thrive and pallor from anaemia. Severe lead poisoning may present with neurological symptoms, including drowsiness, convulsions and coma from lead encephalopathy. Raised intracranial pressure with papilloedema may be present. There is increasing evidence that chronic exposure to relatively low lead levels may be harmful to cognitive development.

The diagnosis is confirmed by elevated blood lead levels. There may be a hypochromic anaemia and basophilic stippling of neutrophils. Radiographs of the

knee or wrist may show 'lead lines', which are dense metaphyseal bands. The source of lead should be identified and removed. Chelating agents are used to form non-toxic lead compounds. In mild cases, D-penicillamine is given orally, and in severe cases sodium calcium edetate (EDTA) is indicated.

Smoking

The harmful effects of smoking are well documented, with a greatly increased risk of developing chronic bronchitis, lung cancer and cardiovascular disease. Unfortunately, many children become regular smokers while still at school. Children should be given appropriate health education, although its effectiveness is limited by the poor example set by the widespread smoking of adults and the difficulties of health education in secondary school age children. When parents or carers smoke, children have been shown to have a higher incidence of bronchitis, asthma, pneumonia and serous otitis media (glue ear). This particularly applies to babies and young children. Maternal smoking places the infant at increased risk of sudden infant death syndrome (SIDS).

Parents' smoking adversely affects their children's health.

Child protection

Children and young people require parents or carers who love, look after, provide shelter and protect them from harm. Unfortunately, this is not the case for all children. Emotional, physical, sexual abuse and neglect of children by parents, carers and others has occurred throughout history. Abuse and neglect seriously decrease the likelihood that a child will reach his or her full potential, although this is not inevitable; some resilient individuals manage despite very difficult circumstances.

Society, including the medical profession, was largely reluctant to accept that child abuse and neglect occurred until the second half of the twentieth century, when attention was drawn by two American paediatricians to the 'battered child'. In many countries, it is now accepted that child abuse and neglect exist and legislation is in place making them a criminal offence.

Following the Second World War, in parallel with the recognition of child abuse, came increasing recognition of human rights. The UN Convention on the Rights of the Child (see Ch. 1) specifically refers to the child's right to be protected from mistreatment, both physical and mental. It gives governments the responsibility to ensure that children are properly cared for and protected from violence, exploitation, abuse and neglect.

In the UK, a series of high profile cases of child abuse, most recently Victoria Climbie and Baby P (Box 7.1), have highlighted the difficulties faced by all professionals involved with the welfare of children, whether social workers, teachers, police, healthcare professionals or others, in recognising and responding appropriately to the alerting signs of child abuse.

Box 7.1 Two high profile cases of child abuse in the UK

Victoria Climbie

- An 8-year-old girl, abused and murdered by her guardians in London in 2000
- Postmortem – 128 separate injuries
- Public outrage led to public inquiry, which resulted in the 'Every Child Matters' initiative, where all organisations involved with providing services to children were instructed to share information and work together to protect children and young people from harm.

Peter Connelly (Baby P)

- The 17-month-old baby died in 2007 following physical abuse throughout his life
- At death – 50 injuries
- He had 35 contacts with healthcare professionals

A report (Laming report) highlighted the need for better information sharing in both cases.

High profile cases of child abuse:

- **can make some doctors frightened to deal with child protection or become oversuspicious from fear of missing cases. However, specialist advice and assistance should always be available.**
- **have led to improved guidelines and procedures**
- **have resulted in better child protection training for all health professionals.**

However, fear of missing child abuse has to be weighed against the damage of falsely accusing parents of abusing their children. This requires sensible judgement, excellent communication with the parents and a professional culture in which any concern that a child is being maltreated can be readily discussed with senior members of the team.

Summary

- Child abuse has existed for centuries but societies have been slow to acknowledge it as a problem.
- Children are vulnerable and cannot protect themselves – the basis of children's rights.
- Protecting children from harm is a key role of parents or carers and all involved with children.
- All healthcare professionals, social workers, teachers, police and others have a duty to ensure that they know what to do if they have concerns that a child may be being abused.

Types of child abuse and neglect

Abuse and neglect are both forms of maltreatment of a child. Somebody may abuse or neglect a child by inflicting harm, or by failing to act to prevent harm. Children may be abused in a family at home or in an institution or community, usually by someone known to them or, rarely, by a stranger. They may be abused by one or more adults or another child or children. Conventionally, child abuse is categorised into:

- Physical abuse
- Emotional abuse
- Sexual abuse
- Neglect
- Fabricated or induced illness (FII).

Physical abuse

Physical abuse may involve hitting, shaking, throwing, poisoning, burning or scalding, drowning, suffocating, or otherwise causing physical harm to a child.

Emotional abuse

Emotional abuse is the persistent emotional maltreatment of a child resulting in severe and persistent adverse effects on the child's emotional development. It may involve conveying to children that they are worthless or unloved, inadequate, or valued only insofar as they meet the needs of another person. It may feature developmentally inappropriate expectations being imposed on children. These may include interactions that are beyond the child's developmental capability, as well as overprotection and abnormal social interaction. It may involve seeing or hearing the ill treatment of another. It may also involve serious bullying that causes children to feel frightened or in danger, or the exploitation or corruption of children. Some level of emotional abuse is involved in all types of maltreatment of a child, although it may occur alone.

Sexual abuse

Sexual abuse involves forcing or enticing a child or young person to take part in sexual activities, including prostitution, whether or not the child is aware of what is happening. The activities may involve physical contact, including penetrative acts such as rape, buggery or oral sex, and/or non-contact activities, such as involving children in looking at or producing pornographic material or watching sexual activities or encouraging children to behave in sexually inappropriate ways.

Neglect

Neglect is the persistent failure to meet a child's basic physical and/or psychological needs, likely to result in the serious impairment of the child's health or development. It may involve a parent or carer failing to provide:

- adequate food and clothing
- shelter, including exclusion from home or abandonment
- protection from physical and emotional harm or danger
- adequate supervision, including the use of inadequate caregivers
- access to appropriate medical care or treatment. It may also include neglect or unresponsiveness to a child's basic emotional needs.

Fabricated or induced illness (FII)

This is a broad term to describe a group of behaviours by parents (or carers), but usually the mother (>80%), which cause harm to children. It fulfils the parents (or carers) own needs. It may consist of:

Verbal fabrication – parents fabricate (i.e. invent) symptoms and signs in the child, telling a false story to healthcare professionals, leading them to believe the child is ill and requires investigation and treatment. Medical and nursing staff are used as the instrument to harm the child through unnecessary interventions, including medication, hospital stays, intrusive tests and surgery. In community settings, the false stories may lead to medication, special diets and a restricted lifestyle or special schools.

Induction of illness may involve:

- suffocation of the child, which may present as an acute life-threatening event (ALTE)
- administration of noxious substances or poisons
- excessive or unnecessary administration of ordinary substances (e.g. excess salt)
- excess or unnecessary use of medication (prescribed for the child or others)
- the use of medically provided portals of entry (such as gastrostomy buttons, central lines).

Organic illness, may coexist with fabricated or induced illness in a child, making the fabrication more difficult to identify. It may manifest as overprotection, imposing unwarranted restrictions or giving treatment that is inappropriate or excessive.

A clue may be that the condition only occurs when the offending parent/carer is present or following a hospital visit. The condition can be extremely difficult to diagnose, but may be suspected if the child has frequent unexplained illnesses and multiple hospital admissions with symptoms that only occur in the carer's presence and are not substantiated by clinical findings. This disorder can be very damaging to the child, as unnecessary investigations and potentially harmful treatment are likely to be given. The child also learns to live with a pattern of illness rather than health. In induced poisoning, the diagnosis is often difficult but can usually be made by identifying the drug in the blood or urine.

Risk factors

Child maltreatment occurs across socioeconomic, religious, cultural, racial and ethnic groups. While no specific causes have been definitively identified that lead a parent or other caregiver to abuse or neglect a child, research has recognised a number of risk factors commonly associated with maltreatment (Box 7.2). Children within families and environments in which these factors exist have a higher probability of experiencing maltreatment. It must be emphasised, however, that while certain factors are often present among

Box 7.2 Risk factors for child abuse

In the child
- Failure to meet parental expectations and aspirations, e.g. disabled, wrong gender
- Resulted from forced, coercive or commercial sex

Parent/carer
- Mental health problems
- Parental indifference, intolerance or over-anxiousness
- Alcohol, drug abuse

In the family
- Step-parents
- Domestic violence
- Multiple/closely spaced births
- Social isolation or a perceived lack of social support
- Young parental age

Environment
- Poverty, poor housing
- Unsavoury neighbourhood.

families where maltreatment occurs, this does not mean that the presence of these factors will *always* result in child abuse and neglect. For example, there is a relationship between poverty and maltreatment, yet most people living in poverty do not harm their children.

Presentation

Child abuse and neglect

Child abuse may present with one or more of:

- Physical symptoms and signs
- Psychological symptoms and signs
- A concerning interaction observed between the child and the parent or carer
- The child may tell someone about the abuse
- The abuse may be observed.

Identification of child abuse in children with disabilities may be more difficult; disability is also a risk factor for child abuse.

In order to diagnose child abuse or neglect, a detailed history and thorough examination are crucial. In most instances where child abuse is considered, seeking advice from colleagues, e.g. more experienced members of the team, paediatric radiologists, paediatric surgeons, is essential.

Factors to consider in the presentation of a physical injury are:

- The child's age and stage of development
- The history given by the child (if they can communicate)
- The plausibility and/or reasonableness of the explanation for the injury
- Any background, e.g. previous child protection concerns, multiple attendances to A&E or general practitioner
- Delay in reporting the injury
- Inconsistent histories from caregivers
- Inappropriate reaction of parents or caregivers who are vague, evasive, unconcerned or excessively distressed or aggressive.

It is often not clear whether an injury is inflicted or non-inflicted. Table 7.4 gives examples of injuries and a guide as to the likelihood that it is due to an inflicted injury. The context and observations of the family are very important in evaluating injuries which may be inflicted.

Key features of bruising
- **The age of a bruise cannot be accurately estimated.**
- **Bruising is hard to detect on children with dark skin.**
- **Mongolian spots can be mistaken for bruises, as they may still be present at several years of age (see Fig. 9.13d).**

Neglect

Consider the possibility of neglect when the child:

- consistently misses important medical appointments
- lacks needed medical or dental care, immunisations or glasses
- seems ravenously hungry
- is dirty
- is wearing inadequate clothing in cold weather
- is abusing alcohol or other drugs
- says there is no one at home to provide care.

Consider the possibility of neglect when the parent or other adult caregiver:

- appears to be indifferent to the child
- seems apathetic or depressed
- behaves irrationally or in a bizarre manner
- is abusing alcohol or other drugs.

Emotional abuse

This damaging form of abuse is the hardest form of abuse to identify in a healthcare setting. Some clues may be found by noting how the parent or caregiver perceives the child. Is the child:

- the 'wrong' gender
- born at a time of parental separation or violence
- seen as unduly 'difficult'?

There may be clues from the behaviour of the child. This depends on the child's age:

- Babies:
  - Apathetic, delayed development, non-demanding
  - Described by the mother as 'spoiled, attention seeking, in control, not loving her'.
- Toddlers and preschool children:
  - Violent, apathetic, fearful.

Table 7.4 Examples of injuries and a guide as to how likely it is due to an inflicted injury

Injury	More likely to be inflicted	May be inflicted or accidental	Less likely or unlikely to be inflicted
Fractures	Any fracture in a non-mobile child (excluding those with a medical condition predisposing to fragile bones) Rib fractures Multiple fractures (unless due to significant accidental trauma, e.g. a road traffic accident) Multiple fractures of different ages	Skull fracture in young child Long bone fractures in a young but mobile child	Fracture in school-age child with witnessed trauma, e.g. fall from swing
Bruises	Bruising in the shape of a hand (Fig. 7.9a) or object Bruises on the neck that look like attempted strangulation Bruises around the wrists or ankles that look like ligature marks Bruise to the buttocks in a child less than 2 years or any age without a good explanation	Bruising to the trunk with a vague history	Bruises on the shins of a mobile child
Burns	Any burn in a child who is not mobile A burn in the shape of an implement – cigarette, iron A 'glove or stocking' burn consistent with forced immersion (Fig. 7.9b)		A burn to the chest of a mobile toddler with splash marks, a history of pulling a cup of hot tea onto himself
Bites	Bruising in the shape of a bite thought unlikely to have been caused by a young child (Fig. 7.9c)		A witnessed biting of one toddler by another

Physical injuries that can be caused by child abuse



(a) (b) (c)

Figure 7.9 **(a)** Bruising from finger trauma to a baby's head. **(b)** Scald with stocking distribution including the soles from forced immersion in hot water. **(c)** A bite mark on an infant's leg. Adult bite marks may be seen in abuse, but bites from other children are not uncommon.

Case History

7.1 Severe child abuse

A 2-month-old boy was brought into A&E by ambulance, with unexplained loss of consciousness. His mother accompanying him appeared to have learning difficulties and could not explain what had happened. His father arrived soon after and said that he had been changing the child's nappy on the floor when suddenly he 'went all floppy and asleep'.

The child was unresponsive and had shallow breathing. His pupils were dilated. He appeared well nourished and was dressed only in a nappy. There were no obvious injuries seen.

Medical management was rapidly instituted. CT head scan showed subdural haemorrhages (Fig. 7.10). A chest X-ray obtained following intubation showed old posterior rib fractures (Fig. 7.11). Subsequent ophthalmological examination showed bilateral retinal haemorrhages (Fig. 7.12).

The child was transferred to an intensive care unit where, sadly, he died. A postmortem skeletal survey showed metaphyseal fractures (Fig. 7.13).

The parents maintained their story, despite the compelling evidence of inflicted head injury and shaking. The case went to the criminal court and both were sentenced on a number of charges.

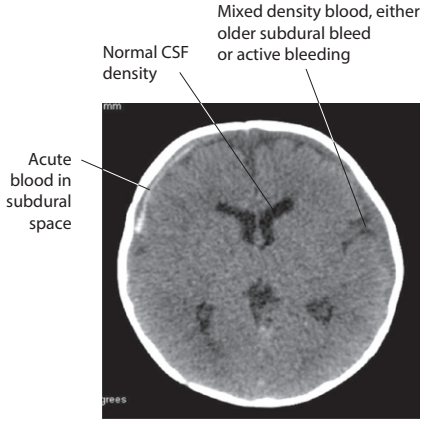


Figure 7.10 Subdural haemorrhage.

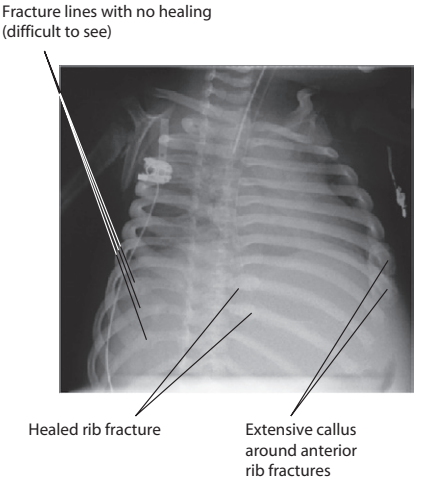


Figure 7.11 Multiple rib fractures of different ages.

Severe physical child abuse resulting in death gains considerable attention from the media but is rare. Many more children suffer permanent injury. Most have been seen previously by health professionals. Early recognition and response to child protection concerns could prevent severe injury.

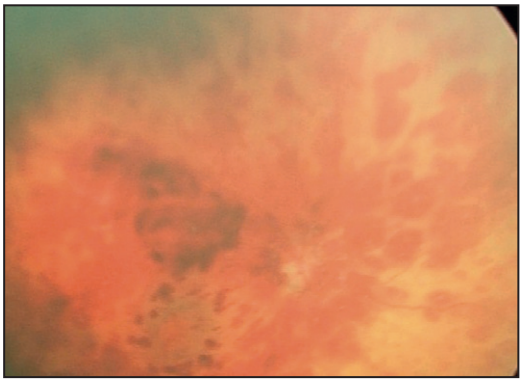


Figure 7.12 Retinal haemorrhages from trauma to the head. (Courtesy of Ms Clare Roberts.)

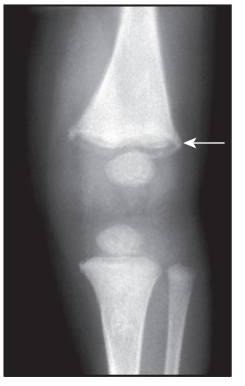


Figure 7.13 Metaphyseal fracture of distal humerus.

- School children:
  - Wetting , soiling, relationship difficulties, non-attendance, anti-social behaviour.
- Adolescents:
  - Self-harm, depression, oppositional, aggressive and delinquent behaviour
  - Being bullied is increasingly recognised as an important form of emotional abuse. Every school should have a written bullying policy which needs to be implemented when necessary.

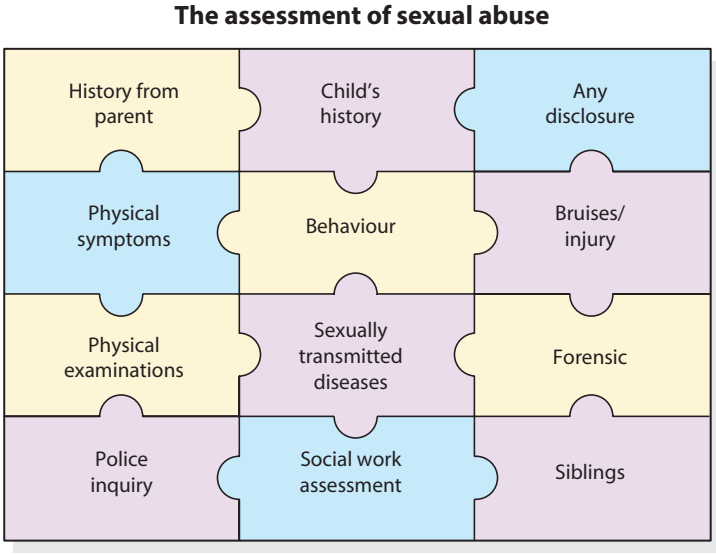


Figure 7.14 The assessment of sexual abuse is like a jigsaw puzzle. Many different pieces of information need to be pieced together to make an informed opinion. (From: *Child Protection Companion,* Royal College of Paediatrics and Child Health, 2005. After Hobbs CJ, Wynne JM, The sexually abused battered child. *Archives of Disease in Childhood* 1990; 65: 423-427. Reproduced with permission from the BMJ Publishing Group.)

Sexual abuse

In suspected sexual abuse, information from different sources needs to be pieced together (Fig. 7.14).

Recognition

The child or young person may:

- tell someone about the abuse
- be identified in pornographic material
- be pregnant (by legal definition this is due to sexual abuse for a girl under the age of 13 years)
- have a sexually transmitted infection with no clear explanation (but some sexually transmitted infections can be passed from the mother to the baby during pregnancy or birth).

Physical symptoms

- Vaginal bleeding, itching, discharge
- Rectal bleeding.

Behavioural symptoms

- Soiling, secondary enuresis
- Self-harm, aggressive or sexualised behaviours, regression, poor school performance.

Signs

There are few clearly diagnostic signs of sexual abuse on examination. This is because sexual abuse of children often comprises touching or kissing or other activities that do not involve significant physical force. Furthermore, the genital area heals very quickly in young children, so signs may be absent even a few days after significant trauma. Forensic material also decays rapidly.

Examination of children suspected of having been sexually abused requires a doctor with specific expertise and training, facilities for photographic documentation, sexually transmitted infection screening and management and, where indicated, forensic testing (Fig. 7.14). Forensic testing of swabs from the child or his/her clothing/bedding may reveal DNA from the sperm of the perpetrator.



Figure 7.15 A thorough medical assessment is required in all children when non-accidental injury is suspected. This girl's large bruise followed what was said to be a minor bump. Non-accidental injury was suspected, but examination showed multiple bruises and petechiae. She had immune thrombocytopenic purpura (ITP).

Investigation

In physical abuse, fractures in young children may not be detectable clinically and X-rays are required to identify them. Bruising overlying a fracture is rarely seen on presentation. A full radiographic skeletal survey with oblique views of the ribs should be performed in all children with suspected physical abuse under 30 months of age. Some lesions may be inconspicuous initially but, if indicated, become evident on a repeat X-ray 1–2 weeks later. Other medical conditions which need to be considered and excluded in suspected child abuse are:

- Bruising – coagulation disorders (Fig. 7.15), Mongolian blue spots on the back or thighs

Case History

7.2 Possible child abuse

Parents brought their 8-month-old daughter into paediatric A&E department. They were worried that she had not been moving her right arm for that day. The family remembered that at the evening meal two evenings before, father was bringing dishes for the family meal to a low corner coffee table in the sitting room. Mother was sitting with baby on her knee, next to the table, trying to control the older siblings, when father had accidentally dropped a heavy serving bowl of food. Mother automatically reached out to try to catch it, dropping the baby in doing so and in the confusion, the serving bowl hit the baby's arm. The baby cried very loudly for about 10 minutes or so but then seemed to settle. The next day she didn't use the right arm but the family thought this was explained by the injury causing a 'strain' as they couldn't see any bruising on the arm. An X-ray showed a fracture of the right radius and ulna (Fig. 7.16).

Child protection concerns

- Baby under 1 year with fracture
- Delayed presentation.

Positive features

- Plausible, consistent story
- Good parent–child interaction observed by medical and nursing staff
- Well-nourished, well-cared-for appearance of baby
- No other injuries on full examination
- Skeletal survey showed no other fractures
- Personal child health record showed regularly weighed, thriving baby up to date with immunisations
- No GP or HV concerns about the family
- Not previously known to local Children's Social Services.

Outcome

- Strategy meeting – no additional concerns identified. Decision – increased Health Visitor contact and parents received advice about safety in the home.

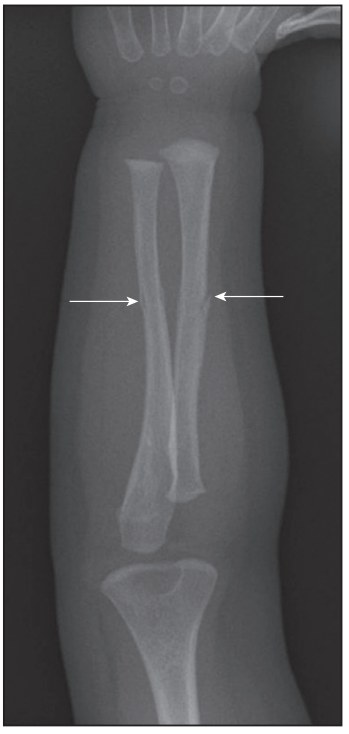


Figure 7.16 X-ray of right arm showing fracture of radius and ulna.

In child protection, conclusive evidence is often not available.

- Fractures – osteogenesis imperfecta, commonly referred to as brittle bone disease. The type commonly involved with unexplained fractures is type I, which is an autosomal dominant disorder, so there may be a family history. Blue sclerae are a key clinical finding and there may be generalised osteoporosis and wormian bones in the skull (extra bones within skull sutures) on skeletal survey.
- Scalds and cigarette burns – may be misinterpreted in children with bullous impetigo or scalded skin syndrome.

Where brain injury is suspected all children require:

- An immediate CT head scan followed later by a MRI head scan
- A skeletal survey to exclude fractures
- An expert ophthalmological examination to identify retinal haemorrhages
- A coagulation screen.

Management

Abused children may present to doctors in the hospital or to medical or nursing staff in the community. They may also be brought for a medical opinion by social services or the police. In all cases, the procedures of the local safeguarding children board should be followed. The medical consultation should be the same as for any medical condition, with a detailed history and full examination. It is usually most productive when this is conducted in a sensitive and concerned way without being accusatory or condemning. Any injuries or medical findings should be carefully noted, measured, recorded and drawn on a body map and photographed (with consent). The height, weight and head circumference (where appropriate) should be recorded and plotted on a centile chart. The interaction between the child and parents should be noted. All notes must be meticulous, dated, timed and signed on each page. Treatment of specific injuries should be instigated and blood tests and X-rays undertaken.

If abuse is suspected or confirmed, a decision needs to be made as to whether the child needs immediate protection from further harm. If this is the case, this may be achieved by admission to hospital, which also allows investigations and multidisciplinary assessment. If sympathetically handled, most parents are willing to accept medical advice for hospital admission for observation and investigation. Occasionally, this is not possible and legal enforcement is required. If medical treatment is not necessary but it is felt to be unsafe for the child to return home, a placement may be found in a foster home.

When dealing with any child suspected of having been abused, the safety of any other siblings or children at home must be considered; the police and/or social services should be alerted to any concerns.

In addition to a detailed medical assessment, evaluation by social workers and other health professionals will be required. A strategy meeting and later a child protection conference may be convened in accordance with local procedures. Members may include social workers, health visitors, police, general practitioner, paediatricians, teachers and lawyers. Parents attend all or part of the case conference. Details of the incident leading to the conference and the family background will be discussed. Good communication and a trusting working relationship between the professionals are vital, as it can be extremely difficult to evaluate the likelihood that injuries were inflicted deliberately and the possible outcome of legal proceedings. The conference will decide:

- whether the child should be provided with a Child Protection Plan and under what category (see Case History 7.1)
- whether there should be an application to the Court to protect the child
- what follow-up is needed.

Why is child protection so difficult? Child protection:

- **goes against the assumption that parents usually have their children's best interests at heart**
- **can involve confronting parents who may be manipulative or aggressive**
- **requires detailed evaluation of the history and examination to identify inconsistencies and interpret subtle findings**
- **depends on genuinely good multi-professional teamwork and respect of colleagues.**

Summary

Child abuse

- Child abuse is the responsibility of all doctors, and must not be avoided or ignored because it raises difficult issues and possible appearance in Court.
- It takes various forms – physical abuse, emotional abuse, sexual abuse, neglect, fabricated or induced illness.
- The interests of the child should be kept uppermost to ensure protection from harm.
- In many instances it is uncertain whether or not the problem is one of child abuse. Good communication with the parents and child is vital.

Further reading

HM Government: *Working Together to Safeguard Children: Every Child Matters*, London, 2006, HM Government.

HM Government: *Children Act*, London, 1989, HM Government.

Laming Report 2003, 2009.

Local Safeguarding Children Board. Policies and Procedures.

NICE: *NICE Guideline – Child Maltreatment*, London, 2009, NICE.

Royal College of Paediatrics and Child Health: *Child Protection Companion and Reader.* UK.

Lancet Series: Child Maltreatment, 2008.

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8

Genetics

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Polygenic, multifactorial or complex inheritance 126
Dysmorphology 127
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Genetic disorders are:

- common, with 2% of live-born babies having a significant congenital malformation and about 5% a genetic disorder
- burdensome to the affected individual, family and society, as many are associated with severe and permanent disability.

There has recently been an unprecedented growth in knowledge about the genetic basis of diseases :

- The Human Genome Project resulted in the first publication of the human genome sequence in 2001.
- It is now estimated that the human genome contains 20 000–25 000 genes, although the function of many of them remains unknown. Greater diversity and complexity at the protein level is achieved by alternative mRNA splicing and post-translational modification of gene products.
- Microarray techniques and high throughput sequencing are increasing the volume and speed of genetic investigations and reducing their costs, leading to a greater understanding of the impact of genetics on health and disease.
- Access to genome browser databases containing DNA sequence and protein structure has greatly enhanced progress in scientific research and the interpretation of clinical test results (Fig. 8.1).
- Genetic databases are available on thousands of multiple congenital anomaly syndromes, on chromosomal variations and disease phenotypes and on all Mendelian disorders.
- Clinical application of these advances is available to families through specialist genetic centres that offer investigation, diagnosis, counselling and antenatal diagnosis for an ever-widening range of disorders.
- Gene-based knowledge is entering mainstream medical and paediatric practice, especially in diagnosis and in therapeutic guidance, such as for the treatment of malignancies.

Genetically determined diseases include those resulting from:

- chromosomal abnormalities
- the action of a single gene (Mendelian disorders)
- unusual genetic mechanisms
- interaction of genetic and environmental factors (polygenic, multifactorial, or complex disorders).

Chromosomal abnormalities

Genes are composed of DNA that is wound around a core of histone proteins and packaged into a succession of supercoils to form the chromosomes. The human chromosome complement was confirmed as 46 in 1956. The chromosomal abnormalities in Down, Klinefelter and Turner syndromes were recognised in 1959 and thousands of chromosome defects have now been documented.

Chromosomal abnormalities are either numerical or structural. They occur in approximately 10% of spermatozoa and 25% of mature oöcytes and are a common cause of early spontaneous miscarriage. The estimated incidence of chromosomal abnormalities in live-born infants is about 1 in 150; they usually cause multiple congenital anomalies and cognitive difficulties. Acquired chromosomal changes play a significant role in carcinogenesis and tumour progression.

Down syndrome (trisomy 21)

This is the most common autosomal trisomy and the most common genetic cause of severe learning difficulties. The incidence (without antenatal screening) in live-born infants is about 1 in 650.

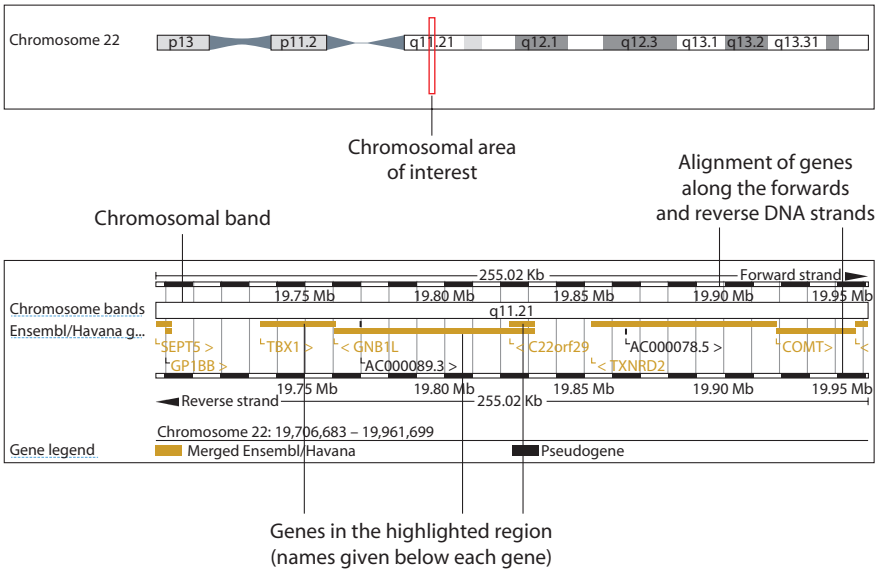


Figure 8.1 Ensembl Genome Browser. The image shows part of chromosome region 22q11, involved in 22q11 deletion syndrome (Di George syndrome). Although only part of the commonly deleted region is shown, the image shows several genes that are deleted in 22q11 deletion syndrome. The online Ensembl browser can be used to 'zoom in' on specific areas, showing the genes present in different chromosome regions, and can also be used to show the gene sequence itself.

Clinical features

Down syndrome is usually suspected at birth because of the baby's facial appearance. Most affected infants are hypotonic and other useful clinical signs include a flat occiput, single palmar creases, incurved fifth finger and wide 'sandal' gap between the big and second toe (Fig. 8.2a–c, Box 8.1). The diagnosis can be difficult to make when relying on clinical signs alone and a suspected diagnosis should be confirmed by a senior paediatrician. Before blood is sent for analysis, parents should be informed that a test for Down syndrome is being performed. The results may take 1–2 days, using rapid FISH (fluorescent in situ hybridisation) techniques. Parents need information about the short- and long-term implications of the diagnosis. They are also likely, at some stage in the future, to appreciate the opportunity to discuss how and why the condition has arisen, the risk of recurrence and the possibility of antenatal diagnosis in future pregnancies.

It is difficult to give a precise long-term prognosis in the neonatal period, as there is individual variation in the degree of learning difficulty and the development of complications. Over 85% of infants with trisomy 21 survive to 1 year of age. Congenital heart disease is present in 30% and, particularly atrioventricular canal defect, is a major cause of early mortality. At least 50% of affected individuals live longer than 50 years. Parents also need to know what assistance is available from both professionals and family support groups. Counselling may be helpful to assist the family to deal with feelings of grief, anger or guilt.

The Child Development Service will provide or coordinate care for the parents. This will include regular review of the child's development and health. Children with Down syndrome are at increased risk of hypothyroidism, impairment of vision and hearing and of atlanto-axial instability.

Cytogenetics

The extra chromosome 21 may result from meiotic non-disjunction, translocation or mosaicism.

Meiotic non-disjunction (94%)

In non-disjunction trisomy 21:

- most cases result from an error at meiosis
- the pair of chromosome 21s fails to separate, so that one gamete has two chromosome 21s and one has none (Fig. 8.3)
- fertilisation of the gamete with two chromosome 21s gives rise to a zygote with trisomy 21
- parental chromosomes do not need to be examined.

The incidence of trisomy 21 due to non-disjunction is related to maternal age (Table 8.1). However, as the proportion of pregnancies in older mothers is small, most affected babies are born to younger mothers. Furthermore, meiotic non-disjunction can occur in spermatogenesis so that the extra 21 can be of paternal origin. All pregnant women are now offered screening tests measuring biochemical markers in blood samples and often also nuchal thickening on ultrasound (thickening of the soft tissues at the back of the neck) to identify an increased risk of Down syndrome in the fetus. When an increased risk is identified, amniocentesis is offered to check the fetal karyotype. After having one child with trisomy 21 due to non-disjunction, the risk of recurrence of Down syndrome is given as 1 in 200 for mothers under the age of 35 years, but remains similar to their age-related population risk for those over the age of 35 years.

Down syndrome



Figure 8.2a Characteristic facies seen in Down syndrome. Her posture is due to hypotonia.



Figure 8.2b Single palmar crease.



Figure 8.2c Pronounced 'sandal' gap between the big and second toe.

Box 8.1 Characteristic clinical manifestations of Down syndrome

Typical craniofacial appearance

- Round face and flat nasal bridge
- Upslanted palpebral fissures
- Epicanthic folds (a fold of skin running across the inner edge of the palpebral fissure)
- Brushfield spots in iris (pigmented spots)
- Small mouth and protruding tongue
- Small ears
- Flat occiput and third fontanelle

Other anomalies

- Short neck
- Single palmar creases, incurved fifth finger and wide 'sandal' gap between toes
- Hypotonia
- Congenital heart defects (40%)
- Duodenal atresia
- Hirschsprung disease

Later medical problems

- Delayed motor milestones
- Moderate to severe learning difficulties
- Small stature
- Increased susceptibility to infections
- Hearing impairment from secretory otitis media
- Visual impairment from cataracts, squints, myopia
- Increased risk of leukaemia and solid tumours
- Risk of atlanto-axial instability
- Increased risk of hypothyroidism and coeliac disease
- Epilepsy
- Alzheimer's disease.

Translocation (5%)

When the extra chromosome 21 is joined onto another chromosome (usually chromosome 14, but occasionally chromosome 15, 22 or 21), this is known as a Robertsonian translocation. This may be present in a phenotypically normal carrier with 45 chromosomes (two being 'joined together') or in someone with Down syndrome and a set of 46 chromosomes but with three copies of chromosome 21 material. In this situation, parental chromosomal analysis is recommended, since one of the parents may well carry the translocation in balanced form (in 25% of cases) (Fig. 8.4).

In translocation Down syndrome:

- The risk of recurrence is 10–15% if the mother is the translocation carrier and about 2.5% if the father is the carrier.
- If a parent carries the rare 21:21 translocation, all the offspring will have Down syndrome.
- If neither parent carries a translocation (75% of cases), the risk of recurrence is <1%.

Mosaicism (1%)

In mosaicism, some of the cells are normal and some have trisomy 21. This usually arises after the formation of the chromosomally normal zygote by non-disjunction at mitosis but can arise by later mitotic non-disjunction in a trisomy 21 conception. The phenotype is sometimes milder in Down syndrome mosaicism.

Inheritance of Down syndrome

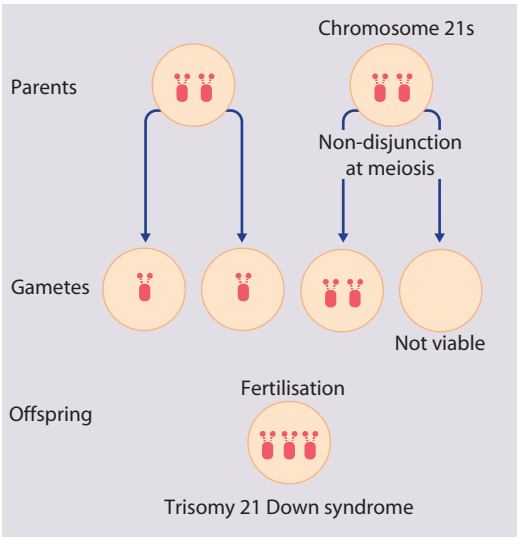


Figure 8.3 Non-disjunction Down syndrome.

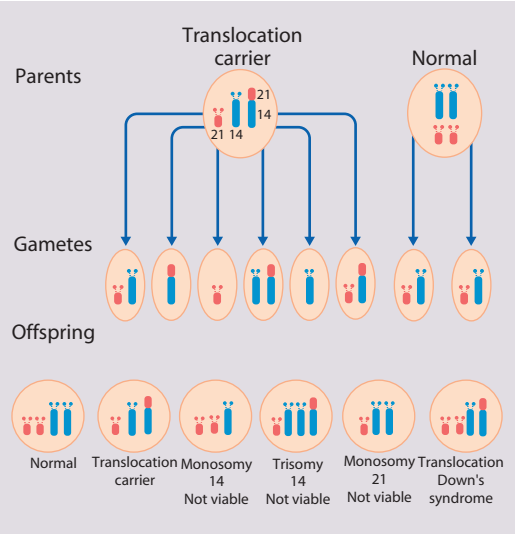


Figure 8.4 Translocation Down syndrome. There is a Robertsonian translocation involving chromosomes 21 and 14, which has been inherited from a parent.

Table 8.1 Risk of Down syndrome (live births) with maternal age at delivery, prior to screening in pregnancy

Maternal age (years)	Risk of Down syndrome
All ages	1 in 650
20	1 in 1530
30	1 in 900
35	1 in 385
37	1 in 240
40	1 in 110
44	1 in 37

Summary

Down syndrome (trisomy 21)

- Natural incidence ~1.5 per 1000 infants
- Cytogenetics – non-disjunction (most common, related to maternal age), translocation (one parent may carry a balanced translocation) or mosaicism (rare)
- Presentation – antenatal screening, prenatal diagnosis or clinical presentation; confirmed on chromosome analysis
- Immediate medical complications – increased risk of duodenal atresia, congenital heart disease
- Clinical manifestations (see Box 8.1).

Edwards syndrome (trisomy 18) and Patau syndrome (trisomy 13)

Although rarer than Down syndrome (1 in 8000 and 1 in 14 000 live births, respectively), particular constellations of severe multiple abnormalities suggest these diagnoses at birth; most affected babies die in infancy (Fig. 8.5, Boxes 8.2 and 8.3). The diagnosis is confirmed by chromosome analysis. Many affected fetuses are detected by ultrasound scan during the second trimester of pregnancy and diagnosis can be confirmed antenatally by amniocentesis and chromosome analysis. Recurrence risk is low, except when the trisomy is due to a balanced chromosome rearrangement in one of the parents.

Turner syndrome (45, X)

Usually (>95%), Turner syndrome results in early miscarriage and is increasingly detected by ultrasound antenatally when fetal oedema of the neck, hands or feet or a cystic hygroma may be identified. In live-born females, the incidence is about 1 in 2500. Figure 8.6 and Box 8.4 show the clinical features of Turner syndrome, although short stature may be the only clinical abnormality in children.

Edwards syndome and Patau syndrome

Box 8.2 Clinical features of Edwards syndrome (trisomy 18)

- Low birthweight
- Prominent occiput
- Small mouth and chin
- Short sternum
- Flexed, overlapping fingers (Fig. 8.5)
- 'Rocker-bottom' feet
- Cardiac and renal malformations.



Figure 8.5 Overlapping of the fingers in Edwards syndrome.

Box 8.3 Clinical features of Patau syndrome (trisomy 13)

- Structural defect of brain
- Scalp defects
- Small eyes (microphthalmia) and other eye defects
- Cleft lip and palate
- Polydactyly
- Cardiac and renal malformations.

Turner syndrome

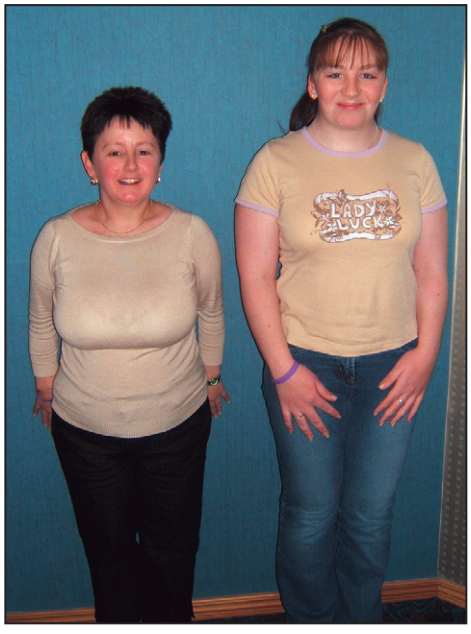


Figure 8.6 Turner syndrome. The woman on the left has marked short stature but no other clinical features; the adolescent female on the right has neck webbing and has received growth hormone and is 150 cm in height.

Box 8.4 Clinical features of Turner syndrome

- Lymphoedema of hands and feet in neonate, which may persist
- Spoon-shaped nails
- Short stature – a cardinal feature
- Neck webbing or thick neck
- Wide carrying angle (cubitus valgus)
- Widely spaced nipples
- Congenital heart defects (particularly coarctation of the aorta)
- Delayed puberty
- Ovarian dysgenesis resulting in infertility, although pregnancy may be possible with in vitro fertilisation (IVF) using donated ova
- Hypothyroidism
- Renal anomalies
- Pigmented moles
- Recurrent otitis media
- Normal intellectual function in most.

Treatment is with:

- Growth hormone therapy
- Oestrogen replacement for development of secondary sexual characteristics at the time of puberty (but infertility persists).

In about 50% of girls with Turner syndrome, there are 45 chromosomes, with only one X chromosome. The other cases have a deletion of the short arm of one X chromosome, an isochromosome that has two long arms but no short arm, or a variety of other structural defects of one of the X chromosomes. The presence of a Y chromosome sequence may increase the risk of gonadoblastoma.

The incidence does not increase with maternal age and risk of recurrence is very low.

Klinefelter syndrome (47, XXY)

This disorder occurs in about 1–2 per 1000 live-born males. For clinical features, see Box 8.5. Recurrence risk is very low.

Reciprocal translocations

An exchange of material between two different chromosomes is called a reciprocal translocation. When this exchange involves no loss or gain of chromosomal

Box 8.5 Clinical features of Klinefelter syndrome

- Infertility – most common presentation
- Hypogonadism with small testes
- Pubertal development may appear normal (some males benefit from testosterone therapy)
- Gynaecomastia in adolescence
- Tall stature
- Intelligence usually in the normal range, but some have educational and psychological problems.

material, the translocation is 'balanced' and usually has no phenotypic effect. Balanced reciprocal translocations are relatively common, occurring in 1 in 500 of the general population. A translocation that appears balanced on conventional chromosome analysis may still involve the loss of a few genes or the disruption of a single gene at one of the chromosomal breakpoints and result in an abnormal phenotype, often including cognitive difficulties. Studying the breakpoints in such individuals has been one way of identifying the location of specific genes.

Unbalanced reciprocal translocations contain an 'incorrect' amount of chromosomal material and often impair both physical and cognitive development, leading to dysmorphic features, congenital malformations, developmental delay and learning difficulties. In a newborn baby, the prognosis is difficult to predict but the effect is usually severe. The parents' chromosomes should be checked to determine whether the abnormality has arisen *de novo*, or as a consequence of a parental rearrangement. Finding a balanced translocation in one parent indicates a recurrence risk for future pregnancies, so that antenatal diagnosis by chorionic villus sampling or amniocentesis should be offered as well as testing relatives who might be carriers.

Deletions

Deletions are another type of structural abnormality. Loss of part of a chromosome usually results in physical abnormalities and cognitive impairment. The deletion may involve loss of the terminal or an interstitial part of a chromosome arm.

An example of a deletion syndrome involves loss of the tip of the short arm of chromosome 5, hence the name 5p- or monosomy 5p. Because affected babies have a high-pitched mewing cry in early infancy, it is also known as *cri du chat* syndrome. Parental chromosomes should be checked to see if one parent carries a balanced chromosomal rearrangement. The clinical severity varies greatly, depending upon the extent of the deletion. It is now possible to specify the genes involved in chromosomal deletions as molecular methods are replacing standard cytogenetic investigations.

An increasing number of syndromes are now known to be due to chromosome deletions too small to be seen by conventional cytogenetic analysis. Submicroscopic deletions can be detected by FISH studies using

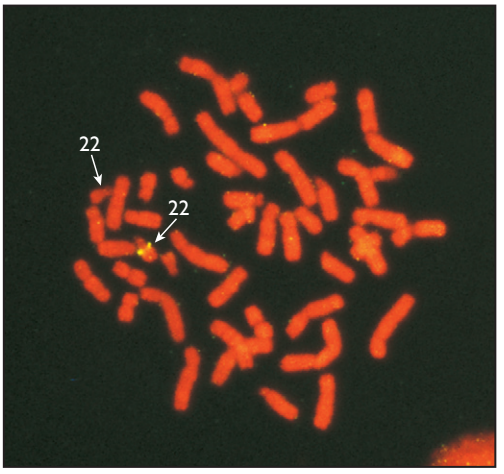


Figure 8.7 Fluorescent in situ hybridisation (FISH) demonstrating a microdeletion on chromosome 22 associated with DiGeorge syndrome. Hybridisation signals are seen on one chromosome 22 but not on the other because of the presence of a deletion. (Courtesy of L. Gaunt, St Mary's Hospital, Manchester.)

DNA probes specific to particular chromosome regions. FISH studies are useful when a specific chromosome deletion is suspected.

DiGeorge syndrome is associated with a deletion of band q11 on chromosome 22 (i.e. 22q11) (Fig. 8.7). Williams syndrome is another example of a microdeletion syndrome due to loss of chromosomal material at band q11 on the long arm of chromosome 7 (i.e. 7q11) (Fig. 8.18, see also Box 8.12).

Mendelian inheritance

Mendelian inheritance, described by Mendel in garden peas in 1866, is the transmission of inherited traits or diseases caused by variation in a single gene in a characteristic pattern. These Mendelian traits or disorders are individually rare but collectively numerous and important: over 6000 have been described. For many disorders, the Mendelian pattern of inheritance is known. If the diagnosis of a condition is uncertain, its pattern of inheritance may be evident on drawing a family tree (pedigree), which is an essential part of genetic evaluation (Fig. 8.8).

Autosomal dominant inheritance

This is the most common mode of Mendelian inheritance (Box 8.6). Autosomal dominant conditions are caused by alterations in only one copy of a gene pair, i.e. the condition occurs in the heterozygous state despite the presence of an intact copy of the relevant gene. Autosomal dominant genes are located on the autosomes (chromosomes 1–22) so males and females are equally affected. Each child from an affected parent has a 1 in 2 (50%) chance of inheriting the abnormal gene (Fig. 8.9a,b). This appears to be straightforward, but complicating factors include the following factors.

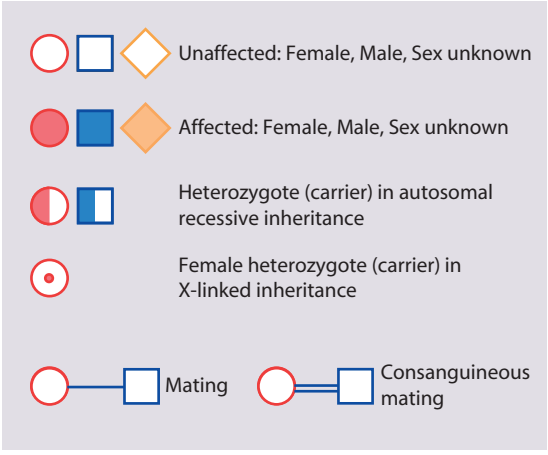


Figure 8.8 Examples of pedigree symbols.

Autosomal dominant inheritance

Box 8.6 Examples of autosomal dominant disorders

- Achondroplasia
- Ehlers–Danlos syndrome
- Familial hypercholesterolaemia
- Huntington disease
- Marfan syndrome
- Myotonic dystrophy
- Neurofibromatosis
- Noonan syndrome
- Osteogenesis imperfecta
- Otosclerosis
- Polyposis coli
- Tuberous sclerosis.

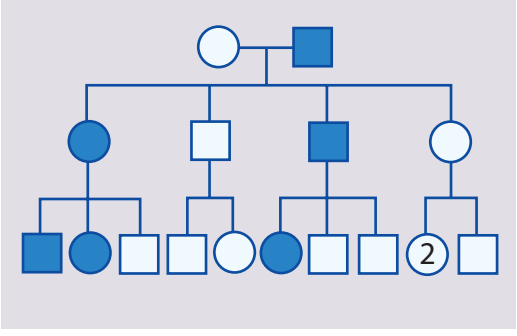


Figure 8.9b Typical pedigree of an autosomal dominant disorder.

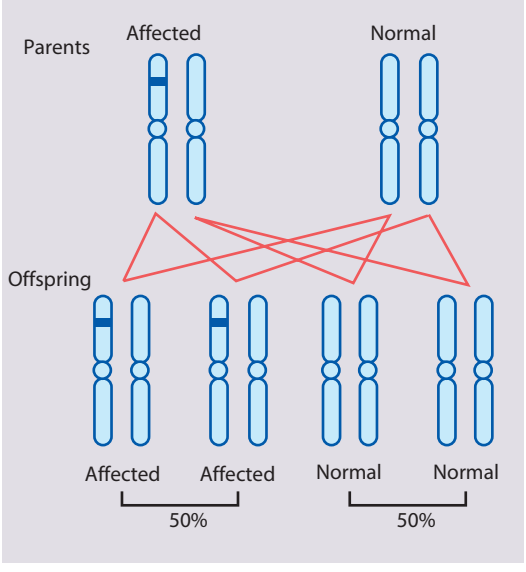


Figure 8.9a Autosomal dominant inheritance.

Summary

Autosomal dominant inheritance

- Most common mode of Mendelian inheritance
- Affected individual carries the abnormal gene on one of a pair of autosomes
- 1 in 2 chance of inheriting the abnormal gene from affected parent, but there may be variation in expression, non-penetrance, no family history (new mutation, parental mosaicism, non-paternity) or homozygosity (rare).

Variation in expression

Within a family, some affected individuals may manifest the disorder mildly and others more severely. This may be the result of variation at other genes, environmental effects, or sheer chance.

Non-penetrance

Refers to the lack of clinical signs and symptoms in an individual who has inherited the abnormal gene. An example of this is otosclerosis, in which only about 40% of gene carriers develop deafness (Fig. 8.10).

No family history of the disorder

It therefore may be due to:

- A new mutation in one of the gametes leading to the conception of the affected person. This is the most common reason for absence of a family history in dominant disorders, e.g. >80% of individuals with achondroplasia have unaffected parents.
- Parental mosaicism – very occasionally a healthy parent harbours the mutation only in some of their cells, e.g. in their gonads. This can account for recurrences of autosomal dominant disorders in siblings born to apparently unaffected parents. It has been described in congenital lethal osteogenesis imperfecta.
- Non-paternity – if the apparent father is not the biological father.

Homozygosity

In the rare situation where both parents are affected by the same autosomal dominant disorder, there is a 1 in 4 risk that a child will be homozygous for the altered gene. This usually causes a more severe phenotype which may be lethal, as with achondroplasia.

Autosomal recessive inheritance

An affected individual is homozygous for the abnormal gene, having inherited an abnormal allele from each parent, both of whom are unaffected heterozygous carriers (Box 8.7). For two carrier parents, the risk of each child, male or female, being affected is 1 in 4 (25%) (Fig. 8.11a,b). All offspring of affected individuals will be carriers.

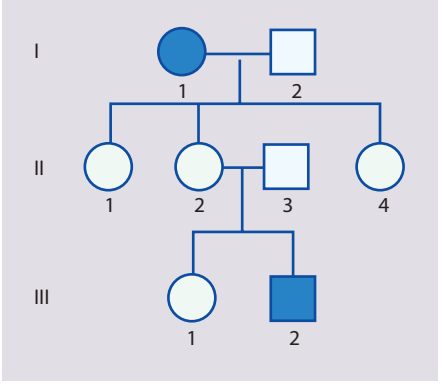


Figure 8.10 Example of non-penetrance. I1 and III2 have otosclerosis. II2 has normal hearing but must have the gene (a new mutation event is most unlikely to arise independently for a second time in the family). The gene is non-penetrant in II2.

Consanguinity

It is thought that we all carry 6–8 abnormal recessive genes. Fortunately, our partners usually carry different ones. Marrying a cousin or another relative increases the chance of both partners carrying the same abnormal autosomal recessive gene. Cousins who marry have a small increase in the risk of having a child with a recessive disorder.

The frequencies of disease alleles at recessive gene loci vary between racial groups. When the gene occurs sufficiently often and the gene or its effect can be detected, population-based carrier screening can be performed and antenatal diagnosis offered for high-risk pregnancies where both parents are carriers. Disorders that can be screened for in this way include sickle cell disease in black Africans and Afro-Americans, the thalassaemias in those from Mediterranean or Asian populations and Tay–Sachs disease in Ashkenazi Jews.

X-linked inheritance

X-linked conditions are caused by alterations in genes found on the X chromosome. These may be inherited as X-linked recessive or X-linked dominant traits but the distinction between these is much less clear than in autosomal traits because of the variable pattern of X chromosome inactivation in females.

In X-linked recessive inheritance (Box 8.8, Fig. 8.12a,b):

- Males are affected
- Female carriers are usually healthy
- Occasionally a female carrier shows mild signs of the disease (manifesting carrier)
- Each son of a female carrier has a 1 in 2 (50%) risk of being affected
- Each daughter of a female carrier has a 1 in 2 (50%) risk of being a carrier
- Daughters of affected males will all be carriers
- Sons of affected males will not be affected, since a man passes a Y chromosome to his son.

The family history may be negative, since new mutations and (gonadal) mosaicism are fairly common. Identification of carrier females in a family requires interpretation of the pedigree, the search for mild clinical manifestations and performing specific biochemical

Summary

X-linked recessive inheritance

- Males are affected; females can be carriers but are usually healthy or have mild disease
- Family history may be negative – new mutations and gonadal mosaicism
- Identifying female carriers is important to be able to provide genetic counselling.

Autosomal recessive inheritance

Box 8.7 Examples of autosomal recessive disorders

- Congenital adrenal hyperplasia
- Cystic fibrosis
- Friedreich ataxia
- Galactosaemia
- Glycogen storage diseases
- Hurler syndrome
- Oculocutaneous albinism
- Phenylketonuria
- Sickle cell disease
- Tay–Sachs disease
- Thalassaemia
- Werdnig–Hoffmann disease (SMA I).

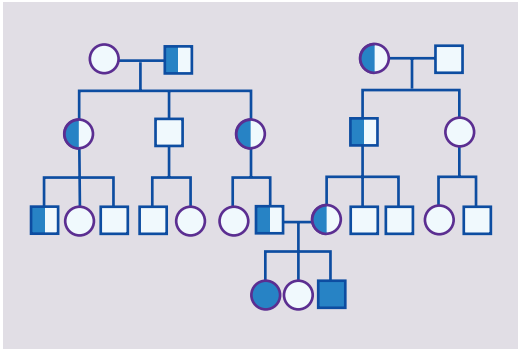


Figure 8.11b Pedigree to show autosomal recessive inheritance.

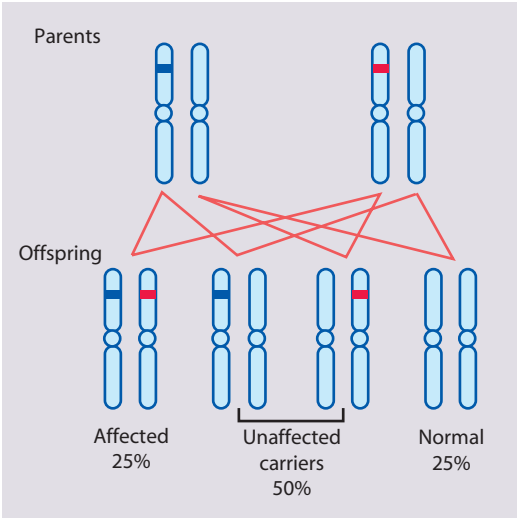


Figure 8.11a Autosomal recessive inheritance.

Summary

Autosomal recessive inheritance

- Affected individuals are homozygous for the abnormal gene; each unaffected parent will be a heterozygous carrier
- Two carrier parents have a 1 in 4 risk of having an affected child
- Risk of these disorders is increased by consanguinity and within specific populations
- Autosomal recessive disorders often affect metabolic pathways, whereas autosomal dominant disorders often affect structural proteins.

or molecular tests. Identifying carriers is important because a female carrier has a 50% risk of having an affected son regardless of who her partner is, and X-linked recessive disorders are often very severe.

X-linked dominant inheritance

X-linked disorders where the mutation has a dominant effect are rare. Both males and females are affected, e.g. hypophosphataemic (vitamin D-resistant) rickets. In some disorders, male lethality is expected and the only affected individuals seen will be female, e.g. incontinentia pigmenti. In others, the condition may affect females because it arises predominantly through mutations at spermatogenesis.

Y-linked inheritance

Y-linked traits are extremely rare. Y-linked inheritance would result in only males being affected, with transmission from an affected father to all his sons. Y-linked genes determine sexual differentiation and spermatogenesis, and mutations are associated with infertility and so are rarely transmitted.

Unusual genetic mechanisms

Trinucleotide repeat expansion mutations

This is a class of unstable mutations caused by unstable expansions of trinucleotide repeat sequences inherited in Mendelian fashion. Fragile X syndrome and myotonic dystrophy were among the first disorders found to be due to such mutations. Other disorders include Huntington disease, spinocerebellar ataxia and Friedreich's ataxia. These disorders follow different patterns of inheritance but share certain unusual properties due to the nature of the underlying mutation. Clinical anticipation is often seen, with the disorders presenting at an earlier age and becoming more severe in successive

X-linked recessive inheritance

Box 8.8 Examples of X-linked recessive disorders

- Colour blindness (red–green)
- Duchenne and Becker muscular dystrophies
- Fragile X syndrome
- Glucose-6-phosphate dehydrogenase (G6PD) deficiency
- Haemophilia A and B
- Hunter syndrome (mucopolysaccharidosis II).

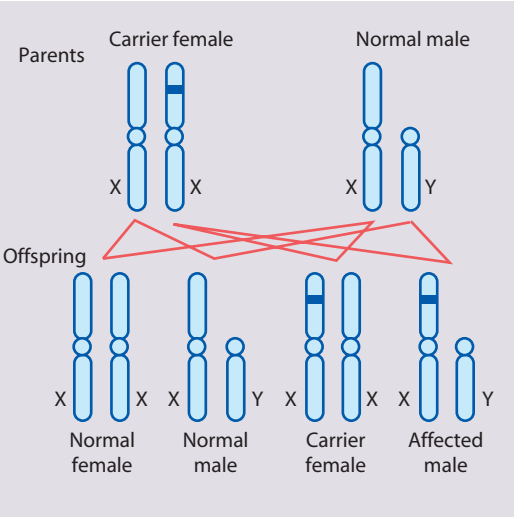
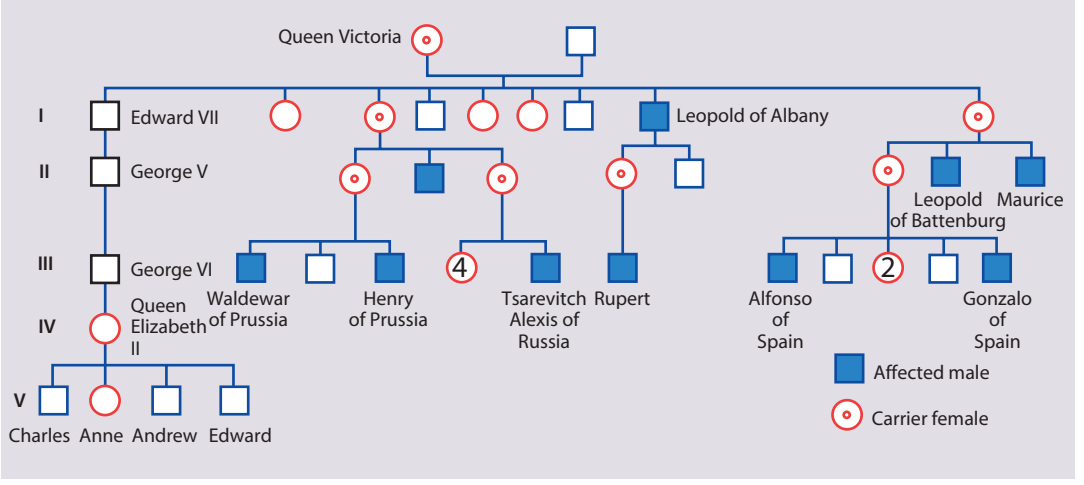


Figure 8.12a X-linked recessive inheritance.

Figure 8.12b Typical pedigree for X-linked recessive inheritance, showing Queen Victoria, a carrier for haemophilia A, and her family. It shows affected males in several generations, related through females, and that affected males do not have affected sons (contrast with autosomal dominant inheritance).



generations of a family as the triplet expands, and with entirely new mutations being exceptionally rare. There are two major categories of triplet repeat disorder, depending upon whether or not the triplet repeat is in the coding sequence of the gene.

Fragile X syndrome

The prevalence of significant learning difficulties in males due to fragile X syndrome is about 1 in 4000 (Fig. 8.13 and Box 8.9). This condition was initially diagnosed on the basis of the appearance of an apparent gap or break (a fragile site) in the distal part of the long arm of the X chromosome. Diagnosis is now achieved by molecular analysis of the CGG trinucleotide repeat expansion in the relevant gene (*FMR1*).

Although it is inherited as an X-linked recessive disorder, a substantial proportion of obligate female carriers have learning difficulties (usually mild to moderate) and around one-fifth of males who inherit the mutation are phenotypically normal but may pass the disorder on to their grandsons through their daughters.

These unusual findings are explained by the nature of the mutation, which occurs in 'pre-mutation' and 'full mutation' forms. The normal copy of the gene contains fewer than 50 copies of the CGG trinucleotide repeat sequence and is stable when transmitted to offspring. Genes with the pre-mutation contain 55–199 copies of the repeat sequence. This expansion causes no intellectual disability in male or female carriers, but is unstable and may become larger during transmission through females. Genes with the full mutation contain more than 200 copies of the repeat sequence. This affects gene function, causing the clinical features of fragile X syndrome in virtually all males and around half of female carriers. These full mutations always arise from expansion of pre-mutations, and never arise directly from normal genes. Hence all mothers of affected males are carriers.

Fragile X



Figure 8.13 A child with fragile X syndrome. At this age, the main physical feature is often the prominent ears.

Box 8.9 Clinical findings in males in fragile X syndrome

- Moderate–severe learning difficulty (IQ 20–80, mean 50)
- Macrocephaly
- Macro-orchidism – postpubertal
- Characteristic facies – long face, large everted ears, prominent mandible and broad forehead, most evident in affected adults
- Other features – mitral valve prolapse, joint laxity, scoliosis, autism, hyperactivity.

Fragile X syndrome is the commonest familial form of learning difficulties and the second most common genetic cause of severe learning difficulties after Down syndrome.

Mitochondrial or cytoplasmic inheritance

Mitochondria are cytoplasmic organelles that function as major energy producers for the cell and contain their own DNA (mtDNA). Each cell contains thousands of copies of the mitochondrial genome. Inherited disorders of the mitochondria may result from mutations in the nuclear genome (the chromosomal genome of the cell nucleus) or in the mitochondria's own genome. In disorders of the mtDNA, the mutation may be present in all or only some of the mitochondria, so that the tissues affected and the severity of the condition can be highly variable. Large deletions of the mtDNA can only be present in a proportion of the mitochondria as they would otherwise be lethal to the cell. Mutations in mtDNA cause overlapping clusters of disease phenotypes (e.g. Leber hereditary optic neuropathy and various mitochondrial myopathies and encephalopathies, MERFF, MELAS, NARP). Mitochondrial DNA mutations show only maternal transmission, since only the egg contributes mitochondria to the zygote.

Imprinting and uniparental disomy

In the past, it was assumed that the activity of a gene is the same regardless of whether it is inherited from the mother or father. It has been shown that the expression of some genes is influenced by the sex of the parent who had transmitted it. This phenomenon is called 'imprinting'. An example involves Prader–Willi syndrome (PWS) (hypotonia, developmental delay, hyperphagia and obesity). The PWS chromosomal region is found at 15q11–13 (i.e. at bands 11–13 on the long arm of chromosome 15). The paternal copy of this chromosomal region has to function for normal development; in its absence, a child will develop PWS. Failure to inherit a functioning maternal copy of this chromosomal region results in an entirely different condition, Angelman syndrome (AS) (causing severe cognitive impairment, a characteristic facial appearance, ataxia and epilepsy), because only the maternal copy of one particular gene in this region is able to function (the paternal copy is inactive because of imprinting). There are two main ways that a child can develop one or other condition:

- De novo *deletion* (Fig. 8.14). Parental chromosomes are normal, and a deletion occurs as a new mutation in the child. If the deletion occurs on the paternal chromosome 15, the child has Prader–Willi syndrome. If the deletion affects the maternal chromosome 15, the child has Angelman syndrome.
- *Uniparental disomy* (Fig. 8.15). This is when a child inherits two copies of a chromosome from one parent and none from the other parent. In Prader–Willi syndrome the affected child has no paternal (but two maternal) copies of chromosome 15q11–13. In Angelman syndrome, the affected child has no maternal (but two paternal) copies of chromosome 15q11–13. This can be detected with DNA analysis.
- There exist other mechanisms that can lead to these conditions.

Imprinting is the unusual property of some genes that express only the copy derived from the parent of a given sex.

Imprinting

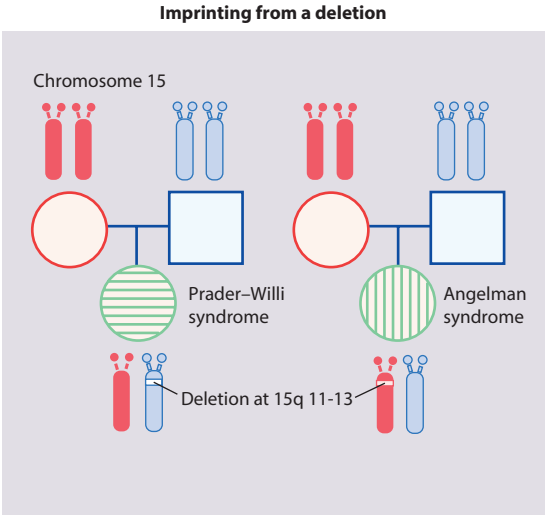


Figure 8.14 Genetic disorder resulting from deletion of an imprinted gene. If the deletion occurs on chromosome 15 inherited from the father, the child has Prader–Willi syndrome. If the deletion occurs on chromosome 15 from the mother, the child has Angelman syndrome.

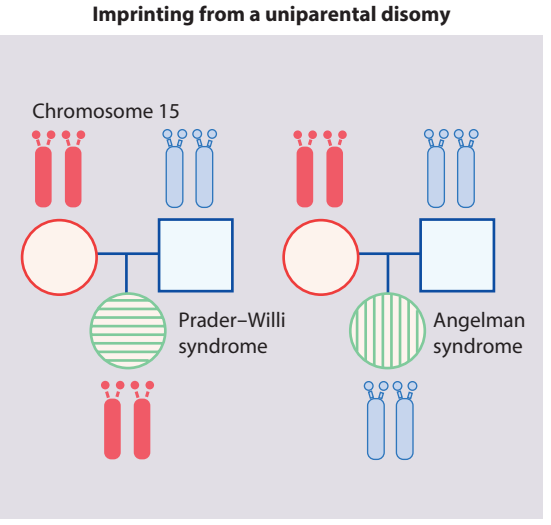


Figure 8.15 Genetic disorder resulting from uniparental disomy affecting imprinted chromosome region. A child who inherits two maternal chromosome 15s will have Prader–Willi syndrome. A child who inherits two paternal chromosome 15s will have Angelman syndrome.

Polygenic, multifactorial or complex inheritance

There is a spectrum in the aetiology of disease, from environmental factors (e.g. trauma) at one end to purely genetic causes (e.g. Mendelian disorder) at the other. Between these two extremes are many disorders which result from the interacting effects of several genes (hence the term polygenic) with or without the influence of environmental or other unknown factors, including chance (multifactorial or complex). The terms are used interchangeably (Box 8.10).

Normal quantitative traits such as height and intelligence are inherited in this fashion, with many relevant influences including genetic constitution, environmental exposures and early life (including intrauterine) experiences. These parameters show a Gaussian (or 'normal') distribution in the population. Similarly, the liability of an individual to develop a disease of multifactorial or polygenic aetiology has a Gaussian distribution. The condition occurs when a certain threshold level of liability is exceeded. Relatives of an affected person show an increased liability due to inheritance of genes conferring susceptibility, and so a greater proportion of them than in the general population will fall beyond the threshold and will manifest the disorder (Fig. 8.16). The risk of recurrence of a polygenic disorder in a family is usually low and is most significant for first-degree relatives. Empirical recurrence risk data are used for genetic counselling. They are derived from family studies that have reported the frequency at which various family members are affected. Factors that increase the risk to relatives are:

- having a more severe form of the disorder, e.g. the risk of recurrence to siblings is greater in bilateral cleft lip and palate than in unilateral cleft lip alone
- close relationship to the affected person, e.g. overall risk to siblings or children is greater than to more distant relatives
- multiple affected family members, e.g. the more siblings already affected, the greater the risk of recurrence
- sex difference in prevalence, with the recurrence risk greater in the more commonly affected sex and if the affected individual is of the less commonly affected sex.

The phenotype (clinical picture) of a disorder may have a heterogeneous (mixed) basis in different families; e.g. hyperlipidaemia leading to atherosclerosis and coronary heart disease can be due to a single gene disorder such as autosomal dominant familial hypercholesterolaemia, but some forms of hyperlipidaemia are polygenic and result from an interaction of the effect of several genes and dietary factors on various lipoproteins.

In some complex disorders, such as Hirschprung disease, the molecular genetic basis and the important contribution of new mutations is becoming clear.

In many multifactorial disorders, the 'environmental factors' remain obscure. Clear exceptions include dietary fat intake and smoking in atherosclerosis, and viral infection in insulin-dependent diabetes mellitus.

Multifactorial, polygenic or complex inheritance

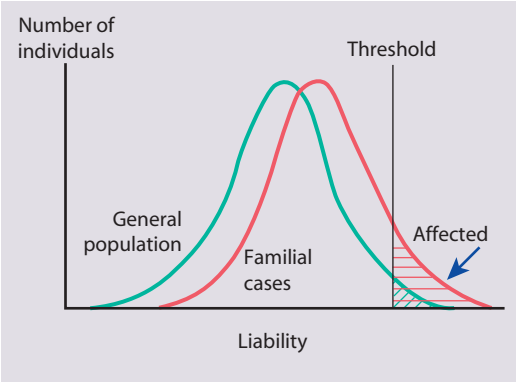


Figure 8.16 Diagram showing the increased liability to a multifactorial disorder in relatives of an affected person.

Box 8.10 Conditions often associated with multifactorial (polygenic, complex) inheritance

Congenital malformations

- Neural tube defects (anencephaly and spina bifida)
- Congenital heart disease
- Cleft lip and palate
- Pyloric stenosis
- Congenital dislocation of the hip
- Talipes equinovarus
- Hypospadias

Childhood

- Atopy (especially asthma and eczema)
- Epilepsy
- Diabetes mellitus type 1 (insulin-dependent diabetes)

Adult life

- Atherosclerosis and coronary artery disease
- Diabetes mellitus type 2
- Alzheimer's disease
- Malignancy (especially the common cancers, e.g. breast and colorectal cancer)
- Hypertension
- Cerebrovascular disease (especially stroke).

For neural tube defects, the risk of recurrence to siblings is lowered from about 4% to 1% or less in future pregnancies if the mother takes folic acid before conception and in the early weeks of pregnancy.

Dysmorphology

The term 'dysmorphology' literally means 'the study of abnormal form' and refers to the assessment of birth defects and unusual physical features that have their origin during embryogenesis.

Pathogenic mechanisms

Malformation

A primary structural defect occurring during the development of a tissue or organ, e.g. spina bifida, cleft lip and palate.

Deformation

Implies an abnormal intrauterine mechanical force that distorts a normally formed structure, e.g. joint contractures or pulmonary hypoplasia due to fetal compression caused by severe oligohydramnios.

Disruption

Involves destruction of a fetal part which initially formed normally, e.g. amniotic membrane rupture may lead to amniotic bands which may cause limb reduction defects.

Dysplasia

Refers to abnormal cellular organisation or function of specific tissue types, e.g. skeletal dysplasias, dysplastic kidney disease.

Clinical classification of birth defects

Single-system defects

These include single congenital malformations, such as spina bifida, which are often multifactorial in nature with fairly low recurrence risks.

Sequence

Refers to a pattern of multiple abnormalities occurring after one initiating defect. Potter syndrome (fetal compression and pulmonary hypoplasia) is an example of a sequence in which all abnormalities may be traced to one original malformation, renal agenesis.

Syndromes recognised by 'Gestalt' (clinical recognition)



Figure 8.17 Noonan syndrome affects males and females. There are some similarities to the phenotype in Turner syndrome, but it is caused by mutation in an autosomal dominant gene and the karyotype is normal.

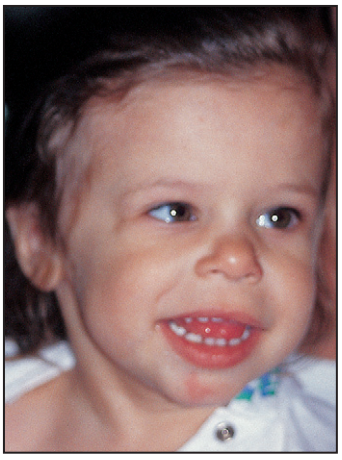


Figure 8.18 Williams syndrome is usually sporadic.

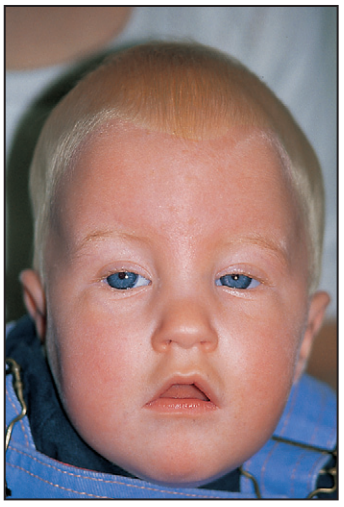


Figure 8.19 Prader–Willi syndrome.

Box 8.11 Clinical features of Noonan syndrome

- Characteristic facies
- Occasional mild learning difficulties
- Short webbed neck with trident hair line
- Pectus excavatum
- Short stature
- Congenital heart disease (especially pulmonary stenosis, atrial septal defect).

Box 8.12 Clinical features of Williams syndrome

- Short stature
- Characteristic facies
- Transient neonatal hypercalcaemia (occasionally)
- Congenital heart disease (supravalvular aortic stenosis)
- Mild to moderate learning difficulties.

Box 8.13 Clinical features of Prader–Willi syndrome

- Characteristic facies
- Hypotonia
- Neonatal feeding difficulties
- Failure to thrive in infancy
- Obesity in later childhood
- Hypogonadism
- Developmental delay
- Learning difficulties.

Association

A group of malformations that occur together more often than expected by chance, but in different combinations from case to case, e.g. VACTERL association (Vertebral anomalies, Anal atresia, Cardiac defects, Tracheo-oEsophageal fistula, Renal anomalies, Limb defects).

Syndrome

When a particular set of multiple anomalies occurs repeatedly in a consistent pattern and there is known or thought to be a common underlying causal mechanism, this is called a 'syndrome'. Multiple malformation syndromes are often associated with moderate or severe cognitive impairment and may be due to:

- chromosomal defects
- a single gene defect (dominant, recessive or sex-linked)
- exposure to teratogens such as alcohol, drugs (especially anticonvulsants such as valproate, carbamazepine and phenytoin) or viral infections during pregnancy
- unknown cause.

Syndrome diagnosis

Although most syndromes are individually rare, recognition of a dysmorphic syndrome is worthwhile as it may give information regarding:

- risk of recurrence
- prognosis
- likely complications which can be sought and perhaps treated successfully if detected early
- the avoidance of unnecessary investigations
- experience and information which parents can share with other affected families through family support groups.

Examples of syndromes recognisable by facial appearance are shown in Figures 8.17–8.19 (see also Boxes 8.11–8.13). The importance and impact of syndrome diagnosis is demonstrated in Case History 8.1. Databases are available to assist with the recognition of thousands of multiple congenital anomaly syndromes (e.g. London Dysmorphology Database (LDDB) & POSSUM).

Summary

Dysmorphology

- comprises birth defects and abnormal clinical features originating during embryogenesis
- may be a malformation, deformation, disruption or dysplasia
- may be classified as a single-system defect, sequence, association or syndrome
- syndromes are recognised by 'Gestalt' (clinical recognition).

Gene-based therapies

The treatment of most genetic disorders is based on conventional therapeutic approaches.

Gene therapy involves the repair, suppression or artificial introduction of genes into genetically abnormal cells with the aim of curing the disease and is at an experimental stage for most genetic conditions being studied. There are still many technical and safety issues to be resolved. Gene therapy has been initiated in adenosine deaminase deficiency (a rare recessive immune disorder), malignant melanoma and cystic fibrosis, and

Case History

8.1 Syndrome diagnosis and genetic counselling

Sean, the second child of healthy parents, was born at term by emergency caesarean section for fetal distress. The pregnancy had been uneventful and no abnormalities were detected on antenatal ultrasound scan. He developed respiratory distress and investigation for a cardiac murmur revealed an interrupted aortic arch and ventricular septal defect that required surgical correction in the neonatal period.

The parents asked about recurrence risk for congenital heart disease and were referred to the genetic clinic. At that time, Sean was thriving and early developmental progress appeared normal. On examination, there were minor dysmorphic features, including a short philtrum, thin upper lip and prominent ears (Fig. 8.20). There was no family history of congenital heart disease or other significant problems and no abnormalities were detected on examination of the parents.

Because of an association between outflow tract abnormalities of the heart and deletions of chromosome 22, cytogenetic analysis was performed using fluorescent in situ hybridisation (FISH). A submicroscopic deletion of the long arm of one chromosome 22 (band 22q11) was detected. Other features of DiGeorge syndrome (hypocalcaemia and T-cell deficiency), which occurs with the same chromosome deletion, were excluded by appropriate tests but could have been important in Sean's medical management.

Parental chromosome analysis showed no deletion at chromosome 22q11 in either parent, indicating a low recurrence risk for future pregnancies since gonadal mosaicism for this deletion is very rare. The older sibling was also normal on testing. Because the parents had normal karyotypes, their own brothers and sisters did not need to be offered tests.

Identification of a 22q11 deletion indicated that other associated problems were likely. Subsequently, Sean required assessment by a multidisciplinary child development team (for developmental delay), that led to educational statementing and recommendation for appropriate placement in a school for children with special educational needs (learning difficulty), input from a clinical psychologist when behavioural problems appeared (ritualistic behaviour and obsessional tendencies), input from speech therapist and plastic surgeon (indistinct speech due to velopharyngeal incompetence) and audiology review (conductive hearing loss due to recurrent otitis media).



Figure 8.20 Sean's facial appearance showing the short philtrum (vertical groove in the upper lip), thin upper lip and prominent ears.

The impact of the diagnosis and its implications was considerable for the family and the parents needed support from a variety of professionals while coming to terms with the various problems as they became apparent. Written information and details of the 22q11 support group were given to the parents. Medical care was coordinated by the paediatrician.

There was the additional worry for the family about a subsequent pregnancy. Fetal echocardiography showed no evidence of congenital heart disease, but invasive tests for cytogenetic analysis were declined because of the low recurrence risk. The baby was born unaffected, with chromosome studies performed on a cord blood sample revealing no abnormality.

some clinical benefit has been reported in a few patients. At present, it is generally accepted that gene therapy should be limited to somatic (not germline) cells, so that the risk of adversely affecting future generations is minimised.

However, other treatments based upon a genetic understanding of disease are being introduced into practice. Two examples are:

- Suppressors of nonsense (stop codon, chain-terminating) mutations, which are under trial in patients with cystic fibrosis and Duchenne muscular dystrophy caused by the appropriate mutations.
- Antisense oligo-DNA molecules that cause the skipping of specific exons in the Duchenne muscular dystrophy gene, i.e. complementary DNA molecules that bind and block exons around the faulty part of the gene (often deletions or duplications in Duchenne muscular dystrophy). This approach can restore the reading frame of the message downstream of the causal deletion; it is hoped that this will result in a milder form of the disease.

Genetic services

In the UK, all health regions have a clinical genetics centre where specialist genetic services are provided by consultants and other medical staff, genetic counsellors and laboratory scientists. Specialist genetic investigations and counselling are provided at the centre and at secondary and primary care. Increased recognition of disorders antenatally has necessitated expansion of perinatal genetic services in addition to paediatric and adult services.

Genetic investigations

For many years genetic investigation relied on determining the karyotype by visualisation under the microscope. This has been transformed by the tremendous advances in molecular testing.

DNA analysis using polymerase chain reaction (PCR) allows rapid analysis on small samples. Its main impact for genetic counselling is:

- Confirmation of a clinical diagnosis of an increasing number of single gene disorders
- Detection of female carriers in X-linked disorders, e.g. Duchenne and Becker muscular dystrophies, haemophilia A and B
- Carrier detection in autosomal recessive disorders, e.g. cystic fibrosis
- Presymptomatic diagnosis in autosomal dominant disorders, e.g. Huntington disease, myotonic dystrophy, familial cancer syndromes
- Antenatal diagnosis of an increasing number of Mendelian conditions.

These are accomplished by the following.

Mutation analysis

For an increasing number of Mendelian disorders, it is possible to directly detect the actual mutation causing the disease. This provides very accurate results for confirmation of diagnosis, and presymptomatic or predictive testing. Identifying the mutation in an affected individual may be very time-consuming, but once this has been done, testing other relatives is usually fairly simple. Examples are:

- *Deletions*: large deletion mutations are common in a variety of disorders including Duchenne and Becker muscular dystrophies, alpha-thalassaemia and 21-hydroxylase deficiency (congenital adrenal hyperplasia).
- *Point mutations and small deletions*: these can be readily identified if the same mutation causes all cases of the disorder, as in sickle cell disease. For most disorders, however, there is a spectrum of mutations. About 78% of cystic fibrosis carriers in the UK possess the ΔF508 mutation, but over 900 other mutations have been identified. Most laboratories test for a certain number of the most common mutations in the population they serve.
- *Trinucleotide repeat expansion mutations*: these are readily tested for because the mutation in a given disease is virtually always at the same site and can be amplified from the same oligo-DNA primers used in the amplification by PCR: the only difference is the size of the repeat sequence, which can be determined from the size of the DNA fragment containing the repeat.

Genetic linkage

If mutation analysis is not available, it may be possible to use DNA sequence variations (markers) located near to, or within, the disease gene to track the inheritance of this gene through a family. This type of analysis requires a suitable family structure and several key members need to be tested to identify appropriate markers before linkage testing can be used in diagnostic, predictive or prenatal testing.

Some of the genetic investigations now available are summarised in Table 8.2.

Genetic counselling

The main aims of genetic counselling are supportive and educational. Genetic counselling aims to support and provide information for individuals, couples and families:

- To understand their situation
- To make their own decisions about managing the disease or risk of disease, including decisions about genetic testing and reproduction
- To adjust to their situation of being affected by or at risk of the genetic condition.

A primary goal of genetic counselling is to provide information to allow for greater autonomy and choice in reproductive decisions and other areas of personal life. Avoiding additional cases of genetic disease in a family may be a consequence of genetic counselling but is not the primary aim. The elements of genetic counselling include:

- Listening to the questions and concerns of the patient, client or family.

Table 8.2 Genetic investigations

Investigation	Application
Cytogenetic analysis – karyotype	Chromosomes stained and visualised under a microscope Detects alterations in chromosome number and structural rearrangements; this method is being replaced by molecular methods such as CGH.
Molecular cytogenetic analysis – FISH (fluorescent in situ hybridisation)	Fluorescent-labelled DNA probes to detect the presence, number and chromosomal location of specific chromosomal sequences Useful for microdeletion syndromes
Microarray comparative genomic hybridisation (aCGH)	Detects chromosomal imbalances using thousands of DNA probes to investigate a whole genome with much greater sensitivity than cytogenetic methods
DNA analysis	Polymerase chain reaction (PCR) to amplify the DNA and determine the sequence of the relevant gene
High throughput DNA sequencing	Rapid sequencing of whole genomes or many loci within the genome
Linkage disequilibrium and genome-wide association studies (GWAS)	Comparing the frequency of combinations of alleles at nearby loci in a given population to identify genetic variants associated with complex diseases

- Establishing the correct diagnosis. This involves detailed history, examination and appropriate investigations that may include chromosome or DNA or other molecular genetic analysis, biochemical tests, X-rays and clinical photographs. Despite extensive investigation, including searching databases, the diagnosis may remain unknown, e.g. in children with learning disability and mild or non-specific dysmorphic features.
- Risk estimation. This requires both diagnostic and pedigree information. Drawing a pedigree of three generations is an essential part of a clinical genetic assessment. The mode of inheritance may be apparent from the pedigree even when the precise diagnosis is not known. In some cases it may not be possible to define a precise recurrence risk and uncertainty may remain, e.g. conditions that only affect one member of a family and are known to follow autosomal dominant inheritance in some families and autosomal recessive inheritance in others (genetic heterogeneity).
- Communication. Information must be presented in an understandable and unbiased way. Families often find written information helpful to refer back to and diagrams are often used to explain patterns of inheritance. The impact of saying 'the recurrence risk is 5% or 1 in 20' may be different from saying 'the chance of an unaffected child is 95% or 19 out of 20', and so both should be presented.
- Discussing options for management and prevention. If there appears to be a risk to offspring, all reproductive options should be discussed. These include not having (any more) children, reducing intended family size, taking the risk and proceeding with pregnancy or having antenatal diagnosis and selective termination of an affected fetus. For some couples donor insemination or ovum donation may be appropriate and for others achieving a pregnancy through IVF (in vitro fertilisation) and preimplantation diagnosis may be possible.

Box 8.14 Influences on decisions regarding options for genetic counselling

- Magnitude of risk
- Perceived severity of disorder
- Availability of treatment
- Person's experience of the disorder
- Family size
- Availability of a safe and reliable prenatal diagnostic test
- Parental cultural, religious or ethical values.

Counselling should be non-directive, but should also assist in the decision-making process (Box 8.14). Information from lay support groups may also be helpful.

Genetic counselling aims to allow parents greater autonomy and choice in reproductive decisions.

Pre-symptomatic (predictive) testing

Children may be referred because they are at increased risk of developing a genetic disorder in childhood or adult life.

If the condition is likely to manifest in childhood (e.g. Duchenne muscular dystrophy) or if there are useful medical interventions available in childhood (e.g. screening by colonoscopy for colorectal tumours in children at risk of familial adenomatosis polyposis coli), then genetic testing is appropriate in childhood.

If the child is at risk of a late-onset and untreatable disorder (e.g. Huntington disease) or if the genetic test result is only relevant to reproductive questions in the future (the child's genetic carrier status), then there is a case for deferring genetic testing until the child can be actively involved in making the decision. These difficult issues are often best handled through a process of genetic counselling supporting open and sustained communication within the family and especially between parents and children.

Pre-symptomatic testing of disorders which manifest in adult life should not be performed until the individual can give informed consent.

Further reading

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E!Ensembl: Available at: http://www.ensembl.org.

Genome browser

GeneTests: http://www.genetests.org.

Review articles on selected genetic disorders.

NCBI MapViewer: Available at: http://www.ncbi.nlm.nih.gov/projects/mapview.

Genome browser

OMIM (Online Mendelian Inheritance in Man): Available at: http://www.ncbi.nlm.nih.gov/omim.

The British, European and American Societies of Human Genetics: Available at: http://www.bshg.org.uk, http://www.eshg.org and http://www.ashg.org.

Websites include professional and policy pages.

Your Genes, Your Health: Available at: http://www.ygyh.org.

Cold Spring Harbor website giving information on some common genetic disorders and links to DNA tutorials.

9

Perinatal medicine

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The term 'perinatal medicine' refers to medical care of the infant before, during and after birth, acknowledging the continuity of fetal and neonatal life. Using modern technology, such as high-resolution ultrasound and DNA analysis, detailed information about the fetus can now be obtained for a large and increasing number of conditions. Close cooperation is important between the professionals involved in the care of the pregnant mother and fetus and those caring for the newborn infant.

Some definitions used in perinatal medicine are:

- Stillbirth – fetus born with no signs of life ≥24 weeks of pregnancy
- Perinatal mortality rate – stillbirths + deaths within the first week per 1000 live births and stillbirths
- Neonatal mortality rate – deaths of liveborn infants within the first 4 weeks after birth per 1000 live births
- Neonate – infant ≤28 days old
- Preterm – gestation <37 weeks of pregnancy
- Term – 37–41 weeks of pregnancy
- Post-term – gestation ≥42 weeks of pregnancy
- Low birthweight (LBW) – <2500 g
- Very low birthweight (VLBW) – <1500 g
- Extremely low birthweight (ELBW) – <1000 g
- Small for gestational age – birthweight <10th centile for gestational age
- Large for gestational age – birthweight >90th centile for gestational age.

Pre-pregnancy care

The better a mother's state of health and nutrition, and the higher her socioeconomic living standard and the quality of healthcare she receives, the greater is the chance of a successful outcome to her pregnancy.

Couples planning to have a baby often ask what they should do to optimise their chances of having a healthy child. They can be informed that for the mother:

- *Smoking* reduces birthweight, which may be of critical importance if born preterm. On average, the babies of smokers weigh 170 g less than those of non-smokers, but the reduction in birthweight is related to the number of cigarettes smoked per day. Smoking is also associated with an increased risk of miscarriage and stillbirth. The infant has a greater risk of sudden infant death syndrome (SIDS).
- *Pre-pregnancy folic acid* supplements reduce the risk of neural tube defects in the fetus. Low-dose folic acid supplementation is recommended for all women planning a pregnancy. A higher dose is recommended for women with, or have a close relative with, a previously affected fetus.
- Any long-term conditions, such as diabetes and epilepsy, must be reviewed and management changed if necessary.
- *Certain medications* such as retinoids, warfarin and sodium valproate must be avoided because of teratogenic effects.
- *Alcohol* ingestion and *drug abuse* (opiates, cocaine) may damage the fetus.
- *Congenital rubella* is preventable by maternal immunisation before pregnancy.
- *Exposure to toxoplasmosis* should be minimised by avoiding eating undercooked meat and by wearing gloves when handling cat litter.
- *Listeria infection* can be acquired from eating unpasteurised dairy products, soft ripened

cheeses, e.g. brie, camembert and blue veined varieties, patés and ready-to-eat poultry, unless thoroughly re-heated.
- *Eating liver* during pregnancy is best avoided as it contains a high concentration of vitamin A.

Any pre-existing maternal medical condition (e.g. hypertension, HIV) or obstetric risk factors for complications of pregnancy or delivery (e.g. recurrent miscarriage or previous preterm delivery) should be identified and treated or monitored. Obesity increases the risk of developing gestational diabetes and pregnancy-induced hypertension.

Couples at increased risk of inherited disorders should receive genetic counselling before pregnancy. They can then be fully informed, decide whether or not to proceed, and consider antenatal diagnosis if available. Pregnancies at increased risk of fetal abnormality include those in which:

- the mother is older (if she is >35 years old, the risk of Down syndrome is >1 in 380), although screening is now available for all mothers
- there is previous congenital abnormality
- there is a family history of an inherited disorder
- the parents are identified as carriers of an autosomal recessive disorder, e.g. thalassaemia
- a parent carries a chromosomal rearrangement
- parents are close blood relatives.

Pre-pregnancy folic acid supplements reduce the risk of neural tube defects in the fetus.

Antenatal diagnosis

Antenatal diagnosis has become available for an increasing number of disorders. Screening tests performed on maternal blood and ultrasound of the fetus are listed in Box 9.1. The main diagnostic techniques for antenatal diagnosis are maternal serum screening, detailed ultrasound scanning, chorionic villus sampling (at >10 weeks of pregnancy) and amniocentesis (>15 weeks) (Fig. 9.1). In some rare conditions, preimplantation genetic diagnosis (PGD) allows genetic analysis of cells from a developing embryo before transfer to the uterus. The structural malformations and other lesions which can be identified on ultrasound are listed in Box 9.2, with an example in Figure 9.2.

Antenatal screening for disorders affecting the mother or fetus allows:

- reassurance where disorders are not detected
- optimal obstetric management of the mother and fetus
- interventions for a limited number of conditions, such as relieving bladder obstruction or draining pleural effusions, to improve perinatal outcome
- counselling and neonatal management to be planned in advance
- the option of termination of pregnancy to be offered for severe disorders affecting the fetus (see Case History 9.1) or compromising maternal health.

Parents require accurate medical advice and counselling to help them with these difficult decisions. Many

Box 9.1 Screening tests for antenatal diagnosis

Maternal blood	Blood group and antibodies – for rhesus and other red cell incompatibilities Hepatitis B Syphilis Rubella HIV infection Neural tube defects – raised maternal serum alphafetoprotein (MSAFP) with spina bifida or anencephaly, but ultrasound alone increasingly used Down syndrome – risk estimate calculated from age, biochemical markers combined with ultrasound screening for nuchal translucency (back of neck). Aim is to detect >75% with <3% false-positive rate. If high risk, fetal chromosome analysis is offered
Ultrasound screening	Gestational age – can be estimated reliably if early in pregnancy Multiple pregnancies – can be identified Structural malformation – 50–70% of major congenital malformations can be detected. If a significant abnormality is suspected, a more detailed scan by a specialist is indicated Fetal growth – can be monitored by serial measurement of abdominal circumference, head circumference and femur length Amniotic fluid volume – oligohydramnios may result from reduced fetal urine production (because of dysplastic or absent kidneys or obstructive uropathy), from prolonged rupture of the membranes or associated with severe intrauterine growth restriction. It may cause pulmonary hypoplasia and limb and facial deformities from pressure on the fetus (Potter syndrome) Polyhydramnios – is associated with maternal diabetes and structural gastrointestinal abnormalities, e.g. atresia in the fetus

Amniocentesis
Chromosome/micro-array and DNA analysis
Fetal infection – PCR

Fetal blood sampling
Fetal haemoglobin for anaemia
Fetal infection serology
Fetal blood transfusion

Chorionic villus sampling
Chromosome/micro-array and DNA analysis
Fetal infection – PCR
Enzyme analysis of inborn error of metabolism

Preimplantation genetic diagnosis (PGD)
In vitro fertilisation allows genetic analysis of cells from developing embryo before transfer to the uterus

Fetoscopy
Minimally invasive surgery, e.g. laser photo-coagulation of communicating vessels in twin–twin transfusion syndrome.

Non-invasive genetic diagnosis – free fetal DNA from maternal blood
Identification of fetal gender and rhesus status

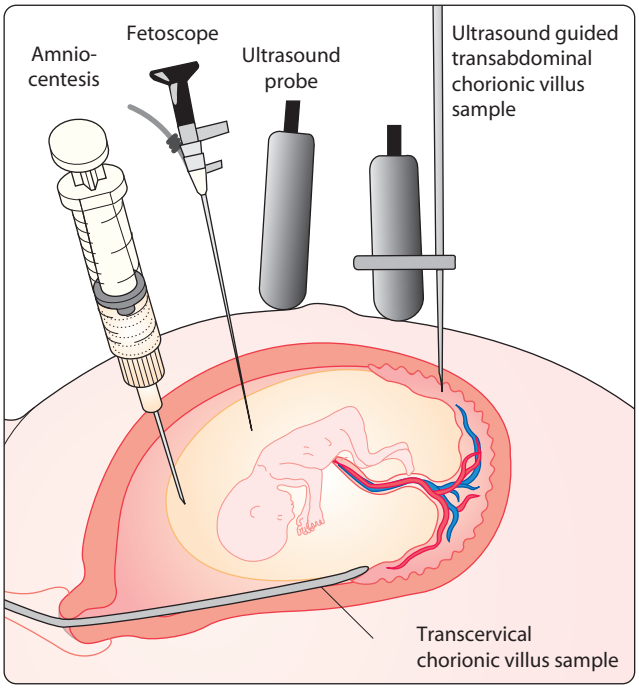


Figure 9.1 Some of the techniques used for antenatal diagnosis.

Box 9.2 Main structural malformations and other lesions detectable by ultrasound

CNS	Anencephaly – always detected
	Spina bifida
	Hydrocephalus, microcephaly, encephalocele
Cardiac	About 50% of severe malformations detected on 'routine' screening, over 90% at specialist centres
Intrathoracica	Diaphragmatic hernia, congenital cystadenomatoid lung malformation (CCAM)
	Oesophageal atresia
Facial	Cleft lip
Gastrointestinal	Bowel obstruction, e.g. duodenal atresia
	Exomphalos and gastroschisis
Genitourinary	Dysplastic or cystic kidneys
	Obstructive disorders of kidneys or urinary tract (hydronephrosis, distended bladder)
Skeletal	Skeletal dysplasias, e.g. achondroplasia and limb reduction deformities
Hydrops	Oedema of the skin, pleural effusions and ascites
Chromosomal	Down syndrome – suspected from thickened back of neck (nuchal translucency), duodenal atresia or an atrioventricular canal defect of the heart. Other chromosomal disorders – from identifying multiple abnormalities

transient or minor structural disorders of the fetus are also detected, which may cause considerable anxiety.

Antenatal diagnosis allows many congenital malformations which used to be diagnosed at birth or during infancy to be identified before birth.

Fetal medicine

The fetus can sometimes be treated by giving medication to the mother. Examples include:

- *Glucocorticoid therapy* before preterm delivery accelerates lung maturity and surfactant production. This has been tested in over 15 randomised trials and markedly reduces